

Distance-Residual Physics-Informed Neural Networks: A Deep Learning Framework for Differential and Partial Differential Inclusions

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Abstract

We introduce *Distance-Residual Physics-Informed Neural Networks* (DR-PINNs), a supervised learning framework for approximating solutions of differential and partial differential inclusions — governing laws in which a differential operator is constrained to lie in a set-valued map rather than equaling a prescribed function. The central idea is a single unifying principle: replace the classical pointwise PDE/ODE residual by the squared distance from the differential operator to the admissible set. This substitution is natural (the distance vanishes precisely when the inclusion is satisfied), geometrically interpretable (it measures the minimum correction needed for the operator to enter the admissible set), and amenable to gradient-based training: for a fixed closed convex admissible set, the squared distance is differentiable with respect to the operator value, with gradient given by the metric projection. When the admissible set also depends on the network state, that dependence must additionally be included by the chain rule. The framework encompasses ordinary differential inclusions and partial differential inclusions with set-valued reaction terms. For both settings we prove *consistency*: any sequence of network approximations whose distance

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residual converges to zero must converge, along a subsequence, to an exact solution of the target inclusion. The proofs combine a priori estimates (Grönwall inequality and parabolic energy estimates), weak compactness arguments (Arzelà–Ascoli and Aubin–Lions), and the lower semicontinuity of normal convex integrands. For admissible sets given as convex hulls of finitely many vertices, projection onto the set reduces to a small convex quadratic program, making the loss efficiently computable inside the training loop. Numerical experiments on benchmark differential inclusions and parabolic inclusions demonstrate the accuracy and flexibility of DR-PINNs.

Keywords: Physics-Informed Neural Networks, distance function, differential inclusions, partial differential inclusions, consistency analysis.

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1. Introduction

Physics-Informed Neural Networks (PINNs) have emerged as a powerful paradigm for approximating solutions of differential equations by embedding the governing physical laws directly into the training loss [1]. Since the foundational work of Raissi, Perdikaris, and Karniadakis, the PINN framework has been applied to a broad spectrum of problems: nonlinear partial differential equations (PDEs) in fluid mechanics, heat transfer, and wave propagation; inverse problems and parameter identification; operator learning; and high-dimensional stochastic differential equations. The approach shares a common structural feature across all of these applications: the governing law takes the form of an equality, either an ordinary differential equation (ODE) of the form $\dot{x}(t) = f(t, x(t))$ or a PDE of the form $\mathcal{D}(u) = f$, and the PINN loss is built from the *pointwise residual* obtained by substituting a neural-network surrogate into that equality.

A class of mathematical models which does not fit within this framework is the one whose governing law takes the *set-valued* form

$$\mathcal{D}(u) \in \mathcal{F}(u),$$

where $\mathcal{F}(u)$ is a multifunction (set-valued map). Concrete instances include:

- *Differential inclusions*: $\dot{x}(t) \in F(t, x(t))$, where $F(t, x)$ is a set of admissible velocities;
- *Parabolic partial differential inclusions*: $\partial_t u - \Delta u \in \Phi(u)$, where $\Phi(u)$ is a set-valued reaction term.

In all of these cases the classical notion of residual breaks down, because there is no single-valued right-hand side against which the differential operator can be compared. A simple geometric observation to overcome this difficulty is the following. Given any set-valued law $\mathcal{D}(u) \in \mathcal{F}(u)$, the quantity

$$\text{dist}(\mathcal{D}(u), \mathcal{F}(u)) := \inf_{v \in \mathcal{F}(u)} \|\mathcal{D}(u) - v\|$$

is always well defined whenever $\mathcal{F}(u)$ is nonempty and closed, and it vanishes if and only if the inclusion is satisfied. This suggests replacing the classical PINN residual by a *distance residual* and training a neural network to minimize the loss

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \text{dist}^2(\mathcal{D}(u_\theta)(z_i), \mathcal{F}(u_\theta)(z_i)),$$

where u_θ is a neural network surrogate of u , and $\{z_i\}_{i=1}^N$ are collocation points. The resulting methodology, which we call *Distance-Residual Physics-Informed Neural Networks* (DR-PINNs), extends the PINN paradigm to genuinely set-valued differential laws, while reducing to the classical formulation when $\mathcal{F}(u) = \{f(u)\}$ is a singleton.

Related work. Distance and projection operators have entered the PINN toolbox for purposes different from ours. Deguchi and Asai [2] use R -function-based approximate distance fields to impose Dirichlet boundary conditions exactly on non-convex domains, so the distance encodes the geometry of the boundary rather than an admissible set for the dynamics. Several recent architectures equip a PINN with a projection layer that maps an unconstrained network output onto a feasible region to guarantee hard-constraint satisfaction; see the KKT-based projection of Iftakher et al. [3] or the integral-conservation projection of Baez et al. [4]. In all of these constructions the projection acts once, on the network output, to enforce a fixed constraint; it does not serve as the training residual, and none of them addresses a genuinely set-valued evolution law.

The work closest in spirit to ours is the series of papers by Wu, Lisser, and Chamoin [5, 6, 7, 8] on solving variational inequalities and complementarity problems via a combination of neurodynamic optimization and PINNs. Their strategy is to first rewrite the variational inequality as an ODE whose right-hand side already contains a metric projection onto the (fixed) feasible set, and then to train a PINN using the ordinary pointwise residual of that auxiliary equation. This differs from our approach in an essential way: in [5, 6, 7, 8] the set-valued aspect is *removed before training*, through a one-time reformulation that presupposes a feasible set fixed in advance; in our approach the inclusion $\dot{x}(t) \in F(t, x(t))$ is left genuinely multivalued and the residual itself is replaced by the distance to the state-dependent set $F(t, x_\theta(t))$. The projection implicit in our loss is therefore recomputed along the current trajectory at every training step.

Finally, it is worth-mentioning that although the theory of differential inclusions is well developed [9, 10, 11], to the best of our knowledge no PINN-type method has previously been proposed for the numerical approximation of solutions of differential inclusions or partial differential inclusions with set-valued reaction terms.

Contributions. The main contributions of this paper are:

- (a) A general *distance-residual principle* (Section 3) that provides a unified method for constructing PINN losses for any set-valued differential

law.

- (b) *Consistency theorems* for both ordinary differential inclusions (Section 4.3) and partial differential inclusions (Section 5), showing that vanishing distance residuals certify admissible solutions.
- (c) A *practical projection algorithm* (Section 4.2) for admissible sets given as convex hulls, based on a small quadratic program combined with the Quickhull algorithm.
- (d) *Numerical experiments* (Section 6) on benchmark differential and partial differential inclusions.

2. Preliminaries

2.1. Set-valued functions

This section collects some basic concepts and notation concerning set-valued functions that will be used throughout the paper. For completeness, we recall the notions of graph and semicontinuity, which play a central role in the subsequent analysis.

Definition 2.1. Let X and Y be nonempty sets. A *set-valued function* (or *multifunction*) from X to Y is a mapping

$$\mathcal{F} : X \rightrightarrows Y,$$

which assigns to each $x \in X$ a subset $\mathcal{F}(x) \subseteq Y$. The *graph* of $\mathcal{F}$ is defined as

$$\text{gr}(\mathcal{F}) := \{(x, y) \in X \times Y : y \in \mathcal{F}(x)\}.$$

For a subset $A \subseteq X$, the image of A under $\mathcal{F}$ is given by

$$\mathcal{F}(A) := \bigcup_{x \in A} \mathcal{F}(x).$$

Definition 2.2. Let X and Y be topological spaces, and $\mathcal{F} : X \rightrightarrows Y$ a set-valued map. We say that $\mathcal{F}$ is *upper semicontinuous* at $x \in X$ whenever for every open set U containing $\mathcal{F}(x)$ there exists a neighborhood V of x such that

$$\mathcal{F}(V) \subset U.$$

Similarly, $\mathcal{F}$ is said to be *lower semicontinuous* at $x \in X$ if for every $y \in \mathcal{F}(x)$ and every neighborhood U of y , there exists a neighborhood V of x such that

$$\mathcal{F}(x') \cap U \neq \emptyset \quad \text{for every } x' \in V.$$

$\mathcal{F}$ is said to be upper semicontinuous (respectively, lower semicontinuous) if it is upper semicontinuous (respectively, lower semicontinuous) at every point $x \in X$. Finally, $\mathcal{F}$ is called *continuous* if it is both upper and lower semicontinuous.

Following [10], our approach emphasizes the representation of a set-valued map through its graph, rather than as a mapping between sets, hence we shall say that a set-valued map satisfies property P if and only if its graph does. For instance, a set-valued map is said to be closed (respectively convex, measurable) if and only if its graph is closed (respectively convex, measurable).

In contrast, $\mathcal{F}$ is said to be *closed-valued*, *convex-valued*, *bounded-valued*, *compact-valued*, or *strict* if for every $x \in X$, the set $\mathcal{F}(x)$ is closed, convex, bounded, compact, or nonempty, respectively.

Proposition 2.3. *Let X and Y be topological spaces and $\mathcal{F} : X \rightrightarrows Y$ a strict set-valued function. If $\mathcal{F}$ is closed, then $\mathcal{F}$ is closed-valued.*

Proof. Fix $x \in X$ and let $\{y_n\}_{n \in \mathbb{N}} \subset \mathcal{F}(x)$ be such that $y_n \rightarrow y$ in Y . Since

$$(x, y_n) \in \text{gr}(\mathcal{F}) \quad \text{and} \quad (x, y_n) \rightarrow (x, y),$$

and $\text{gr}(\mathcal{F})$ is closed, it follows that $(x, y) \in \text{gr}(\mathcal{F})$. Hence $y \in \mathcal{F}(x)$, proving that $\mathcal{F}(x)$ is closed. Since x was arbitrary, $\mathcal{F}$ is closed-valued. $\square$

We conclude by recalling the notions of selection and integral of a set-valued function.

Definition 2.4. Let X and Y be nonempty sets and assume that $\mathcal{F} : X \rightrightarrows Y$ is a strict set-valued function. A selection from $\mathcal{F}$ is a function $f : X \rightarrow Y$ such that $f(x) \in \mathcal{F}(x)$ for every $x \in X$.

Definition 2.5. Let $I \subseteq \mathbb{R}$ be an interval and assume that $\mathcal{F} : I \rightrightarrows \mathbb{R}^d$ is strict and upper semicontinuous. The integral of $\mathcal{F}$ over I is defined as the set

$$\int_I \mathcal{F}(s) ds := \left\{ \int_I f(s) ds : f \text{ is an integrable selection from } \mathcal{F} \right\}.$$

2.2. Projection onto a set

Fix a normed space X . If $C \subseteq X$ is a nonempty closed set, then the infimum

$$\text{dist}(x_0, C) := \inf \{ \|x_0 - x\|_X : x \in C \}, \quad x_0 \in X,$$

is always well defined; it is attained, for instance, whenever X is finite-dimensional or, more generally, whenever C is boundedly compact (in a general infinite-dimensional normed space a nearest point may fail to exist). We say that $z \in C$ is a projection of x_0 onto C whenever $\text{dist}(x_0, C) = \|z - x_0\|_X$. In general there can be several projections of x_0 onto C , that is, there might be more than one point in C closest to x_0 . A fundamental result in convex analysis (see, e.g., [12, Section 8.1]) establishes that, whenever the distance is attained, the projection is unique provided C is closed and convex and the norm $\|\cdot\|_X$ is strictly convex.

In the case $X = \mathbb{R}^n$, we assume throughout this work that the norm is the Euclidean norm $\|\cdot\|_2$, which is strictly convex. To simplify the notation, we write $|\cdot|$ in place of $\|\cdot\|_2$. In this scenario the projection of x_0 onto C is unique and will be denoted by $\Pi_C(x_0)$. In other words,

$$\Pi_C(x_0) := \operatorname{argmin}_{x \in C} |x_0 - x|. \quad (2.1)$$

2.3. Background on Differential and Partial Differential Inclusions

Differential inclusions extend ordinary differential equations by allowing the velocity of the state variable to belong to a set of admissible directions. Let $T > 0$, $x_0 \in \mathbb{R}^d$, and assume that

$$F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$$

is a set-valued map. A *differential inclusion* is a problem of the form

$$\dot{x}(t) \in F(t, x(t)), \quad x(0) = x_0. \quad (2.2)$$

For each pair (t, x) , the set $F(t, x)$ represents the admissible velocities available to the system at time t and state x .

Definition 2.6. A function $x : [0, T] \rightarrow \mathbb{R}^d$ is a *solution* of (2.2) if it is absolutely continuous, satisfies $x(0) = x_0$, and $\dot{x}(t) \in F(t, x(t))$ for almost every $t \in [0, T]$.

Absolute continuity guarantees that $\dot{x}(t)$ exists almost everywhere. Thus, for any solution x , we have the exact representation

$$x(t) = x_0 + \int_0^t \dot{x}(s) ds.$$

However, to translate the inclusion $\dot{x}(t) \in F(t, x(t))$ into an integral characterization in terms of F , regularity conditions are needed: specifically, F must be upper semicontinuous, compact-valued, and convex-valued (see Lemma 2.7 below, whose proof can be found in [9, Lemma 1, p. 99]).

Lemma 2.7. (Integral Representation) *Let $F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ be upper semicontinuous, compact-valued, and convex-valued. Then, a continuous function $x(\cdot)$ is a solution on $[0, T]$ of the differential inclusion*

$$\dot{x}(t) \in F(t, x(t))$$

if and only if for every pair $(t_1, t_2) \in [0, T]$ with $t_1 \leq t_2$,

$$x(t_2) \in x(t_1) + \int_{t_1}^{t_2} F(s, x(s)) ds.$$

The proof of the following result appears in [11, Theorem 10.5, p. 138].

Theorem 2.8 (Basic existence). *Let $F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ be strict, compact-valued, and convex-valued. Assume that $F(t, x)$ is measurable in t , upper semicontinuous in x , and satisfies the linear growth condition*

$$\sup_{v \in F(t, x)} |v| \leq m(t)(1 + |x|)$$

for some $m \in L^1(0, T)$ and each $(t, x) \in [0, T] \times \mathbb{R}^d$. Then, for every $x_0 \in \mathbb{R}^d$, there exists an absolutely continuous solution of (2.2).

Remark 2.9. The assumptions that F is strict, compact-valued and convex-valued are natural: they enable measurable selections, compactness arguments, and fixed-point methods to produce absolutely continuous solutions; see [9, 10, 11]. As we shall see, these are also precisely the assumptions under which the distance function to $F(t, x)$ is well-behaved: it is finite, attained, and (under convexity) differentiable.

Unlike ordinary differential equations, uniqueness of solutions of (2.2) is not expected in general. For a given initial condition x_0 , there may exist many admissible trajectories. The *solution set*

$$\mathcal{S}(x_0) = \{x \in \text{AC}([0, T]; \mathbb{R}^d) : x(0) = x_0, \dot{x}(t) \in F(t, x(t)) \text{ a.e.}\}, \quad (2.3)$$

where $\text{AC}([0, T]; \mathbb{R}^d)$ denotes the collection of absolutely continuous functions from $[0, T]$ into $\mathbb{R}^d$, is naturally set-valued. Consequently, a trained DR-PINN approximates *one* admissible trajectory, not a distinguished one; this interpretation is made precise by the consistency result in Section 4.3.

Next, we consider partial differential inclusions. Although, for the sake of simplicity, in what follows we only address the parabolic case, most of the main ideas and results extend to other types of partial differential inclusions.

Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with $C^{1,1}$ boundary, or a bounded convex domain, let $T > 0$, and set $Q := (0, T) \times \Omega$. Consider the parabolic partial differential inclusion

$$\begin{cases} \partial_t u(t, x) - \Delta u(t, x) \in \Phi(u(t, x)), & (t, x) \in Q, \\ u(0, x) = u_0(x), & x \in \Omega, \\ u(t, x) = 0, & (t, x) \in (0, T) \times \partial\Omega, \end{cases} \quad (2.4)$$

where $\Phi : \mathbb{R} \rightrightarrows \mathbb{R}$ is a set-valued reaction term and $u_0 \in H_0^1(\Omega)$.

The natural functional framework for strong solutions of (2.4) is the *parabolic energy space*

$$\mathcal{W}(0, T) := L^2(0, T; H^2(\Omega) \cap H_0^1(\Omega)) \cap H^1(0, T; L^2(\Omega)),$$

equipped with the norm $\|u\|_{\mathcal{W}}^2 := \|u\|_{L^2(0, T; H^2(\Omega))}^2 + \|\partial_t u\|_{L^2(Q)}^2$. By classical embedding theory and the Aubin–Lions lemma [13],

$$\begin{aligned} \mathcal{W}(0, T) &\hookrightarrow C([0, T]; H_0^1(\Omega)) \quad \text{continuously,} \\ \mathcal{W}(0, T) &\hookrightarrow\hookrightarrow C([0, T]; L^2(\Omega)) \quad \text{compactly.} \end{aligned}$$

The compact embedding into $C([0, T]; L^2(\Omega))$ ensures in particular that the initial condition $u(0, \cdot) = u_0$ is well defined pointwise in time for every $u \in \mathcal{W}(0, T)$.

Definition 2.10. A function $u \in \mathcal{W}(0, T)$ is a *strong solution* of (2.4) if

- (a) $u(0, \cdot) = u_0$ in Ω ,
- (b) $u(t, \cdot)|_{\partial\Omega} = 0$ for a.e. $t \in (0, T)$,
- (c) $\partial_t u(t, x) - \Delta u(t, x) \in \Phi(u(t, x))$ for a.e. $(t, x) \in Q$.

Since $u \in \mathcal{W}(0, T)$ implies $\partial_t u, \Delta u \in L^2(Q)$, the inclusion in (c) is meaningful pointwise almost everywhere.

The *solution set* of (2.4) is

$$\mathcal{S}(u_0) := \{u \in \mathcal{W}(0, T) : u \text{ is a strong solution of (2.4)}\}. \quad (2.5)$$

As in the ODE setting, strong solutions need not be unique, i.e. $\mathcal{S}(u_0)$ may contain multiple elements.

The following existence result guarantees that $\mathcal{S}(u_0)$ is nonempty under the following assumptions on the set-valued reaction term $\Phi : \mathbb{R} \rightrightarrows \mathbb{R}$:

(H1') $\Phi(s)$ is nonempty, closed, and convex for every $s \in \mathbb{R}$.

(H2') There exists a continuous selection $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(s) \in \Phi(s) \quad \forall s \in \mathbb{R}.$$

(H3') There exists a constant $m_0 \geq 0$ such that

$$|f(s)| \leq m_0(1 + |s|) \quad \forall s \in \mathbb{R}.$$

Theorem 2.11 (Existence of strong solutions). *Under assumptions (H1')–(H3'), problem (2.4) admits at least one strong solution, i.e., $\mathcal{S}(u_0) \neq \emptyset$.*

Proof. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the continuous selection provided by assumption (H2'). Consider the semilinear parabolic problem

$$\begin{cases} \partial_t u - \Delta u = f(u) & \text{in } Q, \\ u(0, x) = u_0(x) & \text{in } \Omega, \\ u(t, x) = 0 & \text{on } (0, T) \times \partial\Omega. \end{cases}$$

Since f is continuous and satisfies the linear growth condition

$$|f(s)| \leq m_0(1 + |s|), \quad s \in \mathbb{R},$$

standard existence theory for semilinear parabolic equations (see, e.g., [14, Chapter 7]) yields a strong solution

$$u \in L^2(0, T; H^2(\Omega) \cap H_0^1(\Omega)) \cap H^1(0, T; L^2(\Omega)) = \mathcal{W}(0, T).$$

By construction,

$$\partial_t u(t, x) - \Delta u(t, x) = f(u(t, x)) \in \Phi(u(t, x))$$

for almost every $(t, x) \in Q$. Hence u is a strong solution of (2.4) in the sense of Definition 2.10. Therefore, $\mathcal{S}(u_0) \neq \emptyset$. $\square$

Remark 2.12. The assumptions (H1')–(H3') are not intended to be optimal. They provide a simple and transparent set of conditions ensuring that the solution set $\mathcal{S}(u_0)$ is nonempty, which is sufficient for the consistency analysis developed later.

3. A General Distance-Residual Principle

Assume that X and Y are Banach spaces and consider a differential operator $\mathcal{D} : X \rightarrow Y$. Let $\mathcal{F} : X \rightrightarrows Y$ be strict and closed-valued. We consider the abstract set-valued differential problem

$$\mathcal{D}(u) \in \mathcal{F}(u). \quad (3.1)$$

Special cases of (3.1) include differential inclusions $\dot{x}(t) \in F(t, x(t))$ and parabolic partial differential inclusions $\partial_t u - \Delta u \in \Phi(u)$.

Definition 3.1 (Distance residual). Let X and Y be Banach spaces and assume that $\mathcal{D} : X \rightarrow Y$ is a differential operator. Let $\mathcal{F} : X \rightrightarrows Y$ be strict and closed-valued. For any $u \in X$, the *distance residual* of u with respect to (3.1) is defined as distance

$$R(u) := \text{dist}^2(\mathcal{D}(u), \mathcal{F}(u)) = \inf_{v \in \mathcal{F}(u)} \|\mathcal{D}(u) - v\|_Y^2.$$

The associated DR-PINN loss is

$$\mathcal{L}(\theta) := \frac{1}{N} \sum_{i=1}^N R(u_\theta)(z_i),$$

where u_θ is a neural-network surrogate and $\{z_i\}_{i=1}^N$ are collocation points.

Remark 3.2. Under the hypothesis of the previous definition, the distance residual possesses the following fundamental properties:

- (a) Non-negativity: $R(u) \geq 0$ for all $u \in X$.
- (b) Exactness: if $\mathcal{F}(u)$ is closed, then $R(u) = 0 \iff \mathcal{D}(u) \in \mathcal{F}(u)$.

Moreover, $R(u)^{1/2}$ admits a natural geometric interpretation: it represents the distance from $\mathcal{D}(u)$ to the admissible set $\mathcal{F}(u)$, i.e., the minimum correction needed to satisfy the inclusion at u .

Remark 3.3. Since we are assuming that $\mathcal{F}$ is strict and closed-valued, when $\mathcal{F}(u)$ is convex and Y is a Hilbert space, the projection $P_{\mathcal{F}(u)}$ onto $\mathcal{F}(u)$ is uniquely defined (cf. Section 2.2), and

$$R(u) = \|\mathcal{D}(u) - P_{\mathcal{F}(u)}(\mathcal{D}(u))\|_Y^2.$$

For a *fixed* closed convex set $C \subset Y$, the map $y \mapsto \text{dist}^2(y, C)$ is Fréchet differentiable and

$$\nabla_y \text{dist}^2(y, C) = 2(y - P_C(y)).$$

This is the projection-gradient identity used in the numerical implementations. It is a derivative with respect to the first argument, with the set held fixed; it is not, by itself, a formula for $\nabla R(u)$ when $\mathcal{F}(u)$ depends on u . In that case the state dependence of the admissible set must also be differentiated (or rewritten explicitly, as in the translated-set examples below).

This principle provides a unified framework in which classical PINNs appear as a special case. Indeed, when $\mathcal{F}(u) = \{f(u)\}$ is a singleton, the distance residual reduces to $R(u) = \|\mathcal{D}(u) - f(u)\|_Y^2$, the standard PINN residual.

3.1. A general consistency result

The following result justifies the distance-residual formulation in full generality. It shows that if a sequence of neural-network approximations has vanishing distance residual and converges sufficiently strongly, the limit satisfies the target inclusion.

Lemma 3.4. *Let X and Y be Banach spaces and let $\mathcal{F} : X \rightrightarrows Y$ be a closed set-valued function. Suppose that $x_n \rightarrow x$ in X , $y_n \rightarrow y$ in Y , and*

$$\text{dist}(y_n, \mathcal{F}(x_n)) \rightarrow 0. \quad (3.2)$$

Then $y \in \mathcal{F}(x)$.

Proof. By the definition of the distance to a set, for each $n \in \mathbb{N}$ there exists $a_n \in \mathcal{F}(x_n)$ such that

$$\|y_n - a_n\|_Y \leq \text{dist}(y_n, \mathcal{F}(x_n)) + \frac{1}{n}.$$

From this and (3.2) we deduce that

$$\|y_n - a_n\|_Y \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Consequently, since $y_n \rightarrow y$, we have

$$a_n = y_n - (y_n - a_n) \rightarrow y \quad \text{as } n \rightarrow \infty.$$

Now, since $a_n \in \mathcal{F}(x_n)$, we have $(x_n, a_n) \in \text{gr}(\mathcal{F})$ for all $n \in \mathbb{N}$. Since $\text{gr}(\mathcal{F})$ is closed, $x_n \rightarrow x$, and $a_n \rightarrow y$, it follows that $(x, y) \in \text{gr}(\mathcal{F})$, which gives $y \in \mathcal{F}(x)$. $\square$

Theorem 3.5 (General consistency). *Let X, Y be Banach spaces, assume that $\mathcal{D} : X \rightarrow Y$ is continuous, and let $\mathcal{F} : X \rightrightarrows Y$ be strict and closed (that is, with closed graph). Let $\{u_n\}_{n \in \mathbb{N}} \subset X$ satisfy $u_n \rightarrow u$ in X and*

$$\text{dist}(\mathcal{D}(u_n), \mathcal{F}(u_n)) \rightarrow 0.$$

Then $\mathcal{D}(u) \in \mathcal{F}(u)$.

Proof. Since $\mathcal{D}$ is continuous, $\mathcal{D}(u_n) \rightarrow \mathcal{D}(u)$. The result then follows by applying Lemma 3.4 with $x_n = u_n$ and $y_n = \mathcal{D}(u_n)$. $\square$

Theorem 3.5 applies whenever the functional setting provides strong enough convergence of both u_n and $\mathcal{D}(u_n)$. The concrete consistency results proved in Sections 4.3 and 5 are instances in which this convergence is established using compactness arguments adapted to the ODE and PDE settings, respectively.

4. DR-PINNs for Differential Inclusions

4.1. Formulation

Consider the differential inclusion (2.2) with $F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ satisfying the hypotheses of Theorem 2.8. The classical PINN residual

$$\dot{x}(t) - f(t, x(t))$$

has no analogue when F is set-valued and lacks a single-valued selection. The natural replacement is the distance from the computed velocity to the admissible set:

$$R_F(t; \theta) := \text{dist}^2(\dot{x}_\theta(t), F(t, x_\theta(t))), \quad (4.1)$$

where $x_\theta(t)$ is a neural-network surrogate and $\dot{x}_\theta(t)$ is obtained by automatic differentiation, exactly as in classical PINNs.

Remark 4.1. The assumptions that $F(t, x)$ is nonempty and compact ensure that the distance is finite and attained: for every $v \in \mathbb{R}^d$ there exists $\Pi_{F(t, x)}(v) \in F(t, x)$ with

$$\text{dist}(v, F(t, x)) = |v - \Pi_{F(t, x)}(v)|.$$

Closedness gives the following exactness property:

$$\text{dist}(v, F(t, x)) = 0 \iff v \in F(t, x).$$

When $F(t, x)$ is also convex, the projection $\Pi_{F(t, x)}(v)$ is unique (see Section 2.2). The squared distance becomes

$$\text{dist}^2(v, F(t, x)) = |v - \Pi_{F(t, x)}(v)|^2, \quad (4.2)$$

and its gradient with respect to v is

$$\nabla_v \text{dist}^2(v, F(t, x)) = 2(v - \Pi_{F(t, x)}(v)),$$

which is well-defined and Lipschitz. This identity differentiates only with respect to v , with $F(t, x)$ held fixed. If the admissible set depends on the learned state x_θ , the corresponding chain-rule term must be retained in the implementation. These properties are particularly favorable for gradient-based training.

The DR-PINN loss for the differential inclusion (2.2) is

$$\mathcal{L}_F(\theta) = \frac{1}{N} \sum_{i=1}^N \text{dist}^2(\dot{x}_\theta(t_i), F(t_i, x_\theta(t_i))), \quad (4.3)$$

where $\{t_i\}_{i=1}^N \subset [0, T]$ are collocation points. Boundary and initial conditions are incorporated additively, so that the total loss then takes the form

$$\mathcal{L}(\theta) = \lambda_F \mathcal{L}_F(\theta) + \lambda_0 |x_\theta(0) - x_0|^2,$$

where $\lambda_F, \lambda_0 > 0$ are user-specified weights. Given a neural network $N_\theta(t)$ we set

$$x_\theta(t) = x_0 + t N_\theta(t),$$

which enforces $x_\theta(0) = x_0$, so that only the inclusion residual (4.3) remains in the loss, that is, we can take $\mathcal{L}(\theta) = \mathcal{L}_F(\theta)$, which is more convenient for numerical approximations. Alternatively, the initial constraint can be penalized as a distance to a singleton.

4.2. Computing the Projection onto a General Convex Set

To evaluate the loss (4.3) one must compute the projection onto $F(t, x)$. For simple sets (intervals, balls, boxes) this is explicit. For a large and practically relevant class of admissible sets — those that are, or can be approximated by, convex hulls of finitely many vertices — the projection is computable by a small quadratic program, as we now describe.

Suppose that $F(t, x) = \text{conv}(V(t, x))$, where conv denotes the convex hull and

$$V(t, x) = \{v_1(t, x), \dots, v_K(t, x)\} \subset \mathbb{R}^d.$$

This covers any set of the controlled form $F(t, x) = \{f(t, x, u) : u \in U\}$ with U a polytope (whose image under an affine map is again a polytope), as well as any convex set discretized by a finite point cloud. For every $(t, x) \in [0, T] \times \mathbb{R}^d$, the projection of $v \in \mathbb{R}^d$ onto $F = \text{conv}(V(t, x))$ satisfies

$$\Pi_F(v) = \sum_{k=1}^K \lambda_k v_k(t, x), \quad (\lambda_1, \dots, \lambda_K) = \underset{\lambda \in \Delta_K}{\text{argmin}} \left\| v - \sum_{k=1}^K \lambda_k v_k(t, x) \right\|^2, \quad (4.4)$$

where $\Delta_K = \{\lambda \in \mathbb{R}^K : \lambda_k \geq 0, \sum_k \lambda_k = 1\}$ is the probability simplex. Problem (4.4) is a small convex quadratic program with K variables, one equality constraint, and non-negativity bounds. It can be solved efficiently, even inside the training loop, by standard active-set or Frank–Wolfe-type methods.

This reformulation shifts the difficulty from “compute the projection onto a general convex set” to “produce a vertex set $V(t, x)$ whose convex hull approximates $F(t, x)$ ”. When $V(t, x)$ is redundant, it is advantageous to first reduce it to its extreme points using a convex-hull algorithm such as Quickhull [15]. Quickhull computes the extreme points $\tilde{V}(t, x) \subset V(t, x)$ in output-sensitive time and satisfies $\text{conv}(\tilde{V}(t, x)) = \text{conv}(V(t, x)) = F(t, x)$, while typically having far fewer points, which keeps the quadratic program (4.4) small.

Two regimes are worth distinguishing:

- (a) $F(t, x)$ fixed, or depending only on t . The extreme-point set $\tilde{V}(t)$ can be precomputed *offline* on a grid of times. During training only the small quadratic program (4.4) is solved online.
- (b) $F(t, x)$ depends on the state $x_\theta(t)$. If $V(t, x) = f(t, x) + A(t, x)V_0$ for a fixed reference set V_0 , a convex-hull reduction $\tilde{V}_0$ of V_0 can still be computed once offline, since $\text{conv}(AV_0) = A \text{conv}(V_0)$ for any fixed linear map A . Only the reduced set needs to be transformed at each collocation point during training.

In summary, Quickhull acts as a preprocessing step that reduces a large or redundant point cloud to its extreme points, keeping the downstream quadratic program small. Combined with (4.4), this gives a practical route to evaluate $\text{dist}^2(v, F(t, x))$ — and hence the loss (4.3) — for convex sets well beyond the simple examples treated numerically below.

4.3. Consistency

We now prove that the distance-residual formulation is mathematically consistent: any sequence of trajectories whose residual tends to zero must,

along a subsequence, converge to an exact solution of (2.2). This is the analogue, for inclusions, of the consistency statement underlying the classical PINN paradigm, with the crucial difference that no uniqueness — and hence no Grönwall-type error bound — can be expected in general; the correct and strongest available statement is a compactness/lower semicontinuity result in the spirit of the direct method of the calculus of variations.

We work under the hypotheses of Theorem 2.8, with the harmless strengthening $m \in L^2(0, T)$:

(H1) F is strict, compact-valued, and convex-valued.

(H2) $F(t, x)$ is measurable in t and upper semicontinuous in x .

(H3) There exists $m \in L^2(0, T)$, $m \geq 0$, such that

$$\sup_{v \in F(t, x)} |v| \leq m(t)(1 + |x|) \quad \text{for a.e. } (t, x) \in [0, T] \times \mathbb{R}^d.$$

Definition 4.2. Define the admissible space

$$\mathcal{A} := \left\{ x \in W^{1,2}(0, T; \mathbb{R}^d) : x(0) = x_0 \right\},$$

where $W^{1,2}(0, T; \mathbb{R}^d)$ is the Sobolev space of absolutely continuous functions with weak derivative in $L^2(0, T; \mathbb{R}^d)$. For $x \in \mathcal{A}$, define the *distance-residual functional*

$$\mathcal{J}(x) := \int_0^T \text{dist}^2(\dot{x}(t), F(t, x(t))) dt. \quad (4.5)$$

Note that $\mathcal{A}$ is exactly the class of trajectories realizable by the hard-constraint parametrization $x_\theta(t) = x_0 + tN_\theta(t)$.

Lemma 4.3. *Let $F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ be a set-valued function satisfying hypothesis (H1) and (H2). Consider the function $g : [0, T] \times \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ defined as*

$$g(t, x, v) := \text{dist}^2(v, F(t, x)).$$

Then, g is a normal convex integrand, that is:

- (a) $g(t, x, \cdot)$ is convex in v for a.e. $t \in [0, T]$ and $x \in \mathbb{R}^d$.
- (b) $g(t, \cdot, \cdot)$ is jointly lower semicontinuous in (x, v) for a.e. $t \in [0, T]$.
- (c) $g(\cdot, x, v)$ is measurable in t for a.e. $x, v \in \mathbb{R}^d$.

Proof. We start with the proof of (a). Fix $x \in \mathbb{R}^d$. For $v_1, v_2 \in \mathbb{R}^d$ and $\lambda \in [0, 1]$, since $F(t, x)$ is compact and convex, the projections onto $F(t, x)$ exist and are unique:

$$w_1 := \Pi_{F(t, x)}(v_1), \quad w_2 := \Pi_{F(t, x)}(v_2).$$

As $F(t, x)$ is convex, $\bar{w} := \lambda w_1 + (1 - \lambda)w_2 \in F(t, x)$. Consequently,

$$\begin{aligned} \text{dist}^2(\lambda v_1 + (1 - \lambda)v_2, F(t, x)) &\leq |\lambda v_1 + (1 - \lambda)v_2 - \bar{w}|^2 \\ &= |\lambda(v_1 - w_1) + (1 - \lambda)(v_2 - w_2)|^2. \end{aligned} \quad (4.6)$$

On the other hand, since the squared Euclidean norm is convex,

$$|\lambda(v_1 - w_1) + (1 - \lambda)(v_2 - w_2)|^2 \leq \lambda|v_1 - w_1|^2 + (1 - \lambda)|v_2 - w_2|^2. \quad (4.7)$$

Gathering (4.6) and (4.7),

$$\begin{aligned} \text{dist}^2(\lambda v_1 + (1 - \lambda)v_2, F(t, x)) &\leq \lambda|v_1 - w_1|^2 + (1 - \lambda)|v_2 - w_2|^2 \\ &= \lambda \text{dist}^2(v_1, F(t, x)) + (1 - \lambda) \text{dist}^2(v_2, F(t, x)). \end{aligned}$$

We conclude that $g(t, x, \cdot)$ is convex in v .

Next, we address the proof of (b). We start by proving lower semicontinuity in x , with $v \in \mathbb{R}^d$ fixed. Suppose, for the sake of contradiction, that there is a sequence $\{x_n\}_{n=1}^\infty$ with $x_n \rightarrow x \in \mathbb{R}^d$ and

$$\liminf_{n \rightarrow \infty} \text{dist}(v, F(t, x_n)) = L < d := \text{dist}(v, F(t, x)).$$

Choose $d' \in (L, d)$. Since $\text{dist}(v, F(t, x)) = d > d'$, all points in $F(t, x)$ lie outside the closed ball $\bar{B}(v, d')$, that is,

$$F(t, x) \subset U := \mathbb{R}^d \setminus \bar{B}(v, d'),$$

where U is open. By upper semicontinuity of $F(t, \cdot)$ at x (hypothesis H2), there exists a neighborhood V of x such that

$$F(t, x') \subset U \quad \text{for all } x' \in V.$$

This means that $|w - v| > d'$ for all $w \in F(t, x')$ and $x' \in V$. Hence,

$$\text{dist}(v, F(t, x')) \geq d' \quad \text{for all } x' \in V. \quad (4.8)$$

Since $x_n \rightarrow x$ and V is a neighborhood of x , there exists $N \geq 1$ such $x_n \in V$ whenever $n > N$. From this and (4.8), we have

$$\text{dist}(v, F(t, x_n)) \geq d' \quad \text{for all } n > N.$$

This contradicts $\text{dist}(v, F(t, x_n)) \rightarrow L < d'$ and finishes the proof of the lower semicontinuity in x whenever $v \in \mathbb{R}^d$ is fixed.

Next, for a fixed $x \in \mathbb{R}^d$, it is easy to check that the map $v \mapsto \text{dist}(v, F(t, x))$ is 1-Lipschitz continuous in v , that is,

$$|\text{dist}(v, F(t, x)) - \text{dist}(v', F(t, x))| \leq |v - v'| \quad \text{for each } v, v' \in \mathbb{R}^d. \quad (4.9)$$

Fix $\{x_n\}_{n=1}^\infty \subset \mathbb{R}^d$ and $\{v_n\}_{n=1}^\infty \subset \mathbb{R}^d$ with $(x_n, v_n) \rightarrow (x, v)$ as $n \rightarrow \infty$. From (4.9), we have

$$\text{dist}(v_n, F(t, x_n)) \geq \text{dist}(v, F(t, x_n)) - |v_n - v| \quad \text{for all } n \in \mathbb{N}. \quad (4.10)$$

Taking the $\liminf_{n \rightarrow \infty}$ on both sides, the term $|v_n - v|$ vanishes and the lower semicontinuity in x for the first term on the right-hand side gives

$$\liminf_{n \rightarrow \infty} \text{dist}(v_n, F(t, x_n)) \geq \text{dist}(v, F(t, x)). \quad (4.11)$$

The proof of (b) is completed by squaring both sides.

Finally, the claim in (c) follows directly from hypothesis (H2). $\square$

Theorem 4.4 (Consistency of the distance residual). *Let $F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ be a set-valued function satisfying hypothesis (H1), (H2), and (H3). Consider the differential inclusion (2.2), along with its solution set $\mathcal{S}(x_0)$ defined as in (2.3). Then:*

- (a) (Exactness) *For every $x \in \mathcal{A}$, $\mathcal{J}(x) = 0$ if and only if $x \in \mathcal{S}(x_0)$.*
- (b) (Asymptotic consistency) *Let $\{x_n\}_{n=1}^\infty \subset \mathcal{A}$ with $\mathcal{J}(x_n) \rightarrow 0$. Then there exist a subsequence (not relabeled) and $x^* \in \mathcal{A}$ such that $x_n \rightarrow x^*$ uniformly on $[0, T]$, $\dot{x}_n \rightharpoonup \dot{x}^*$ weakly in $L^2(0, T; \mathbb{R}^d)$, and $x^* \in \mathcal{S}(x_0)$.*

Proof. Part (a) follows immediately from the exactness property of the distance. To prove (b) we start by setting $r_n(t) := \text{dist}(\dot{x}_n(t), F(t, x_n(t)))$ for every $n \in \mathbb{N}$ so that

$$\int_0^T r_n^2(t) dt = \mathcal{J}(x_n) =: \varepsilon_n \rightarrow 0.$$

Let $\varepsilon_0 := \sup_n \varepsilon_n < \infty$. Since $F(t, x_n(t))$ is compact and convex, the projection

$$w_n(t) = \Pi_{F(t, x_n(t))}(\dot{x}_n(t)) \in F(t, x_n(t))$$

is uniquely defined and the minimum is attained, that is,

$$|\dot{x}_n(t) - w_n(t)| = \min_{w \in F(t, x_n(t))} |\dot{x}_n(t) - w| = r_n(t).$$

By the triangle inequality and (H3), for a.e. $t \in [0, T]$,

$$|\dot{x}_n(t)| \leq |\dot{x}_n(t) - w_n(t)| + |w_n(t)| = r_n(t) + m(t)(1 + |x_n(t)|). \quad (4.12)$$

On the other hand, applying Cauchy–Schwarz inequality to $r_n(t)$, we obtain

$$\int_0^t r_n(s) ds \leq \left(\int_0^T ds \right)^{1/2} \left(\int_0^T r_n^2(s) ds \right)^{1/2} \leq \sqrt{T\varepsilon_n} \leq \sqrt{T\varepsilon_0} =: R_0.$$

Then, integrating (4.12) we deduce that

$$\begin{aligned} |x_n(t)| &\leq |x_0| + |x_n(t) - x_0| \leq |x_0| + \int_0^t |\dot{x}_n(s)| ds \\ &\leq |x_0| + R_0 + \int_0^t m(s)(1 + |x_n(s)|) ds. \end{aligned}$$

From this and Grönwall's inequality we obtain

$$1 + |x_n(t)| \leq (1 + |x_0| + R_0) e^{\|m\|_{L^1(0,T)}} := C_0 \quad (4.13)$$

for every $t \in [0, T]$ and $n \in \mathbb{N}$. Substituting (4.13) back into (4.12), squaring, and integrating,

$$\int_0^T |\dot{x}_n(t)|^2 dt \leq \int_0^T r_n^2(t) dt + 2C_0 \int_0^T r_n(t)m(t) dt + C_0^2 \int_0^T m^2(t) dt.$$

The second term in the right-hand side can be handled by applying Cauchy–Schwarz to obtain

$$\int_0^T r_n(t)m(t) dt \leq \left(\int_0^T r_n^2(t) dt \right)^{1/2} \cdot \left(\int_0^T m^2(t) dt \right)^{1/2} = \sqrt{\varepsilon_n} \|m\|_{L^2(0,T)}.$$

Therefore,

$$\int_0^T |\dot{x}_n(t)|^2 dt \leq \varepsilon_n + 2C_0\sqrt{\varepsilon_n} \|m\|_{L^2(0,T)} + C_0^2 \|m\|_{L^2(0,T)}^2.$$

Since $\varepsilon_0 := \sup_n \varepsilon_n < \infty$, for every $n \in \mathbb{N}$,

$$\|\dot{x}_n\|_{L^2(0,T)}^2 \leq \varepsilon_0 + 2C_0\sqrt{\varepsilon_0} \|m\|_{L^2(0,T)} + C_0^2 \|m\|_{L^2(0,T)}^2 =: C_1. \quad (4.14)$$

This shows that the sequence $\{\dot{x}_n\}_{n=1}^\infty$ is bounded in $L^2(0, T; \mathbb{R}^d)$. In particular, from (4.13) and (4.14) we deduce that the sequence $\{x_n\}_{n \in \mathbb{N}}$ is bounded

in the Sobolev space $W^{1,2}(0, T; \mathbb{R}^d)$, that is, $\|x_n\|_{W^{1,2}(0, T)} \leq C_W$ for every $n \in \mathbb{N}$.

Since, by (4.14), the sequence $\{\dot{x}_n\}_{n \in \mathbb{N}}$ is bounded in the Hilbert space $L^2(0, T; \mathbb{R}^d)$, which is reflexive, [16, Theorem 3.18] provides a subsequence (not relabeled) and $v^* \in L^2(0, T; \mathbb{R}^d)$ such that

$$\dot{x}_n \rightharpoonup v^* \quad \text{weakly in } L^2(0, T; \mathbb{R}^d).$$

By Cauchy–Schwarz inequality,

$$|x_n(t) - x_n(s)| = \left| \int_s^t \dot{x}_n(\tau) d\tau \right| \leq \|\dot{x}_n\|_{L^2(s, t)} \sqrt{|t - s|} \leq \|\dot{x}_n\|_{L^2(0, T)} \sqrt{|t - s|}.$$

From this and (4.14), we have

$$|x_n(t) - x_n(s)| \leq \sqrt{C_1} \sqrt{|t - s|},$$

hence x_n is uniformly equicontinuous. Since x_n is also uniformly bounded as seen from (4.13), we may therefore apply the Arzelà–Ascoli theorem, which guarantees the existence of a subsequence, still denoted by $\{x_n\}_{n=1}^\infty$, such that

$$x_n \rightarrow x^* \quad \text{uniformly on } [0, T].$$

Then, passing to the limit in $x_n(t) = x_0 + \int_0^t \dot{x}_n(s) ds$ (the integral converges by the weak convergence $\dot{x}_n \rightharpoonup v^*$ in L^2 , tested against $\mathbf{1}_{[0, t]}$) gives

$$x^*(t) = x_0 + \int_0^t v^*(s) ds$$

hence $x^* \in \mathcal{A}$ and $\dot{x}^* = v^*$ a.e.

Finally, set $g(t, x, v) := \text{dist}^2(v, F(t, x))$. By Lemma 4.3, g is a normal convex integrand. The classical lower semicontinuity theorem for integral functionals with normal convex integrands ([17, Theorem 1]; see also [18] and [19, Chapter VIII]), applied with $x_n \rightarrow x^*$ uniformly and $\dot{x}_n \rightharpoonup \dot{x}^*$ weakly in $L^2(0, T; \mathbb{R}^d)$, yields

$$\mathcal{J}(x^*) \leq \liminf_{n \rightarrow \infty} \mathcal{J}(x_n) = 0.$$

As $\mathcal{J}$ is non-negative, we conclude that $\mathcal{J}(x^*) = 0$. By part (a), this means that $x^* \in \mathcal{S}(x_0)$. $\square$

Remark 4.5. Theorem 4.4 should be read together with the non-uniqueness phenomenon discussed in Section 2.3: the theorem does *not* guarantee that the learned trajectory approximates any particular target in $\mathcal{S}(x_0)$, only that it approximates *some* admissible solution. In view of Theorem 2.8, this is the strongest statement available for a set-valued right-hand side.

5. DR-PINNs for Partial Differential Inclusions

Consider now the parabolic partial differential inclusion (2.4).

5.1. The Distance-Based PDE Loss

The classical PDE residual $\partial_t u_\theta - \Delta u_\theta - f(u_\theta)$ has no analogue when Φ is set-valued. The natural replacement is

$$\mathcal{L}_{\text{incl}}(\theta) = \frac{1}{N_r} \sum_{i=1}^{N_r} \text{dist}^2\left(\partial_t u_\theta(t_i, x_i) - \Delta u_\theta(t_i, x_i), \Phi(u_\theta(t_i, x_i))\right), \quad (5.1)$$

where $\{(t_i, x_i)\}_{i=1}^{N_r}$ are collocation points in Q . As in the case of differential inclusions, $\mathcal{L}_{\text{incl}}(\theta) = 0$ if and only if the inclusion is satisfied at every collocation point. The full DR-PINN loss incorporates initial and boundary data:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{incl}}(\theta) + \lambda_0 \mathcal{L}_0(\theta) + \lambda_b \mathcal{L}_b(\theta),$$

where

$$\mathcal{L}_0(\theta) = \frac{1}{N_0} \sum_{j=1}^{N_0} |u_\theta(0, x_j^0) - u_0(x_j^0)|^2, \quad \mathcal{L}_b(\theta) = \frac{1}{N_b} \sum_{k=1}^{N_b} |u_\theta(t_k^b, x_k^b)|^2.$$

5.2. Consistency

For the consistency analysis, we assume that the set-valued reaction term $\Phi : \mathbb{R} \rightrightarrows \mathbb{R}$ satisfies:

(P1) $\Phi(s)$ is nonempty, closed, and convex for every $s \in \mathbb{R}$.

(P2) Φ is upper semicontinuous.

(P3) There exists a constant $m_0 \geq 0$ such that

$$\sup_{r \in \Phi(s)} |r| \leq m_0(1 + |s|) \quad \forall s \in \mathbb{R}.$$

Definition 5.1. Let

$$\mathcal{A} := \{u \in \mathcal{W}(0, T) : u(0, \cdot) = u_0 \text{ in } \Omega, u(t, \cdot)|_{\partial\Omega} = 0 \ \forall t \in [0, T]\}.$$

For $u \in \mathcal{A}$, define the *continuous distance-residual functional*

$$\mathcal{J}(u) := \iint_Q \text{dist}^2(\partial_t u - \Delta u, \Phi(u)) \, dx \, dt.$$

Lemma 5.2 (Normal convex integrand). *Let $\Phi : \mathbb{R} \rightrightarrows \mathbb{R}$ satisfy hypotheses (P1) and (P2). Define $g : \mathbb{R} \times \mathbb{R} \rightarrow [0, \infty)$ by*

$$g(s, z) := \text{dist}^2(z, \Phi(s)).$$

Then g is a normal convex integrand, that is:

- (a) $g(s, \cdot)$ is convex for every $s \in \mathbb{R}$;
- (b) g is jointly lower semicontinuous on $\mathbb{R} \times \mathbb{R}$;
- (c) g is Borel measurable; consequently, for every pair of measurable functions $u, v : Q \rightarrow \mathbb{R}$, the composition $(t, x) \mapsto g(u(t, x), v(t, x))$ is Lebesgue measurable on Q .

Proof. We prove (a), (b), and (c) in turn.

Proof of (a). Fix $s \in \mathbb{R}$. Since $\Phi(s)$ is closed and convex by (P1), and the Euclidean norm on $\mathbb{R}$ is strictly convex, the projection $\Pi_{\Phi(s)}$ is well defined and single-valued. For $z_1, z_2 \in \mathbb{R}$ and $\lambda \in [0, 1]$, set $w_1 := \Pi_{\Phi(s)}(z_1)$, $w_2 := \Pi_{\Phi(s)}(z_2)$. By convexity of $\Phi(s)$, $w := \lambda w_1 + (1 - \lambda)w_2 \in \Phi(s)$, hence

$$\text{dist}^2(\lambda z_1 + (1 - \lambda)z_2, \Phi(s)) \leq |\lambda z_1 + (1 - \lambda)z_2 - w|^2 = |\lambda(z_1 - w_1) + (1 - \lambda)(z_2 - w_2)|^2.$$

By convexity of $t \mapsto t^2$,

$$|\lambda(z_1 - w_1) + (1 - \lambda)(z_2 - w_2)|^2 \leq \lambda(z_1 - w_1)^2 + (1 - \lambda)(z_2 - w_2)^2,$$

and combining the two inequalities gives

$$g(s, \lambda z_1 + (1 - \lambda)z_2) \leq \lambda g(s, z_1) + (1 - \lambda)g(s, z_2).$$

Proof of (b). We first show that $s \mapsto \text{dist}(z, \Phi(s))$ is lower semicontinuous for fixed $z \in \mathbb{R}$. Suppose, for contradiction, that $s_n \rightarrow s$ and

$$\liminf_{n \rightarrow \infty} \text{dist}(z, \Phi(s_n)) = L < d := \text{dist}(z, \Phi(s)).$$

Choose $d' \in (L, d)$. Since $d > d'$, we have $\Phi(s) \subset U := \mathbb{R} \setminus [z - d', z + d']$, an open set. By the upper semicontinuity of Φ at s (hypothesis (P2)), there exists a neighborhood V of s such that $\Phi(s') \subset U$ for all $s' \in V$. Hence $|w - z| > d'$ for every $w \in \Phi(s')$, $s' \in V$, so that

$$\text{dist}(z, \Phi(s')) \geq d' \quad \text{for all } s' \in V.$$

Since $s_n \rightarrow s$, there exists $N \in \mathbb{N}$ such that $s_n \in V$ for all $n > N$, whence $\text{dist}(z, \Phi(s_n)) \geq d'$ for $n > N$. This contradicts $\liminf_n \text{dist}(z, \Phi(s_n)) = L < d'$, proving lower semicontinuity in s .

On the other hand, for fixed $s \in \mathbb{R}$, the map $z \mapsto \text{dist}(z, \Phi(s))$ is 1-Lipschitz:

$$|\text{dist}(z, \Phi(s)) - \text{dist}(z', \Phi(s))| \leq |z - z'| \quad \text{for all } z, z' \in \mathbb{R}.$$

Let $(s_n, z_n) \rightarrow (s, z)$. Then

$$\text{dist}(z_n, \Phi(s_n)) \geq \text{dist}(z, \Phi(s_n)) - |z_n - z|.$$

Taking $\liminf_{n \rightarrow \infty}$ on both sides, the term $|z_n - z|$ vanishes and the lower semicontinuity in s established above gives

$$\liminf_{n \rightarrow \infty} \text{dist}(z_n, \Phi(s_n)) \geq \text{dist}(z, \Phi(s)).$$

Squaring both (nonnegative) sides yields $\liminf_n g(s_n, z_n) \geq g(s, z)$, which proves (b).

Proof of (c). A jointly lower semicontinuous function on a metric space has closed sublevel sets $\{(s, z) : g(s, z) \leq c\}$ for every $c \in \mathbb{R}$, hence g is Borel measurable on $\mathbb{R} \times \mathbb{R}$. If $u, v : Q \rightarrow \mathbb{R}$ are Lebesgue measurable, then $(t, x) \mapsto (u(t, x), v(t, x))$ is measurable from Q into $\mathbb{R}^2$, and the composition of a Borel measurable function with a Lebesgue measurable function is Lebesgue measurable. Therefore $(t, x) \mapsto g(u(t, x), v(t, x))$ is measurable on Q . $\square$

Theorem 5.3 (Consistency: parabolic case). *Assume that $\Phi : \mathbb{R} \rightrightarrows \mathbb{R}$ satisfies (P1)–(P3). Then:*

(a) Exactness. *For every $u \in \mathcal{A}$,*

$$\mathcal{J}(u) = 0 \quad \Longleftrightarrow \quad \partial_t u(t, x) - \Delta u(t, x) \in \Phi(u(t, x)) \quad \text{for a.e. } (t, x) \in Q.$$

(b) Asymptotic consistency. *Let $(u_n)_{n \in \mathbb{N}} \subset \mathcal{A}$ satisfy*

$$\mathcal{J}(u_n) \rightarrow 0.$$

Then there exist a subsequence, not relabeled, and $u^ \in \mathcal{A}$ such that*

$$\begin{aligned} u_n &\rightharpoonup u^* \quad \text{weakly in } \mathcal{W}(0, T), \\ u_n &\rightarrow u^* \quad \text{strongly in } C([0, T]; L^2(\Omega)), \end{aligned}$$

and

$$u^* \in \mathcal{S}(u_0).$$

Proof. Part (a) follows directly from the fact that $\Phi(u(t, x))$ is closed for almost every $(t, x) \in Q$.

We now prove part (b). Set

$$z_n := \partial_t u_n - \Delta u_n$$

and

$$\rho_n(t, x) := \text{dist}(z_n(t, x), \Phi(u_n(t, x))).$$

Then

$$\|\rho_n\|_{L^2(Q)}^2 = \mathcal{J}(u_n) \longrightarrow 0. \quad (5.2)$$

By (P1), the set $\Phi(s)$ is nonempty, closed, and convex for every $s \in \mathbb{R}$. Hence the metric projection onto $\Phi(s)$ is well defined and unique. Define

$$w_n(t, x) := \Pi_{\Phi(u_n(t, x))}(z_n(t, x)).$$

The map w_n is measurable and satisfies

$$w_n(t, x) \in \Phi(u_n(t, x)) \quad \text{for a.e. } (t, x) \in Q,$$

together with

$$|z_n(t, x) - w_n(t, x)| = \rho_n(t, x).$$

Setting

$$r_n := z_n - w_n,$$

we obtain

$$\partial_t u_n - \Delta u_n = w_n + r_n \quad \text{in } Q, \quad (5.3)$$

and, by (5.2),

$$\|r_n\|_{L^2(Q)}^2 = \mathcal{J}(u_n) \longrightarrow 0. \quad (5.4)$$

We first derive an L^2 energy estimate. Testing (5.3) with $u_n(t)$ and using the homogeneous Dirichlet boundary condition, we obtain, for almost every $t \in (0, T)$,

$$\frac{1}{2} \frac{d}{dt} \|u_n(t)\|_{L^2(\Omega)}^2 + \|\nabla u_n(t)\|_{L^2(\Omega)}^2 = \int_{\Omega} (w_n(t, x) + r_n(t, x)) u_n(t, x) \, dx. \quad (5.5)$$

By (P3),

$$|w_n(t, x)| \leq m_0(1 + |u_n(t, x)|) \quad \text{for a.e. } (t, x) \in Q. \quad (5.6)$$

Consequently,

$$\|w_n(t)\|_{L^2(\Omega)} \leq m_0 \left(|\Omega|^{1/2} + \|u_n(t)\|_{L^2(\Omega)} \right). \quad (5.7)$$

Using the Cauchy–Schwarz and Young inequalities in (5.5), together with (5.7), we find a constant $C > 0$, independent of n , such that

$$\frac{d}{dt} \|u_n(t)\|_{L^2(\Omega)}^2 + 2\|\nabla u_n(t)\|_{L^2(\Omega)}^2 \leq C \left(1 + \|u_n(t)\|_{L^2(\Omega)}^2\right) + \|r_n(t)\|_{L^2(\Omega)}^2. \quad (5.8)$$

Since $u_n(0) = u_0$ for every n , Gronwall's inequality and (5.4) yield

$$\sup_{n \in \mathbb{N}} \|u_n\|_{L^\infty(0,T;L^2(\Omega))} < \infty. \quad (5.9)$$

Integrating (5.8) over $(0, T)$ also gives

$$\sup_{n \in \mathbb{N}} \|u_n\|_{L^2(0,T;H_0^1(\Omega))} < \infty.$$

From (5.6) and (5.9), it follows that

$$\sup_{n \in \mathbb{N}} \|w_n\|_{L^2(Q)} < \infty.$$

Together with (5.4), this implies

$$\sup_{n \in \mathbb{N}} \|w_n + r_n\|_{L^2(Q)} < \infty. \quad (5.10)$$

We now apply the standard L^2 regularity estimate for the Dirichlet heat equation to (5.3). Since $u_0 \in H_0^1(\Omega)$ and $\partial\Omega$ is smooth ($C^{1,1}$ regularity is enough; convexity of Ω also suffices, which covers the square domain used in Section 6.3), there exists a constant $C_{\text{par}} > 0$, independent of n , such that

$$\|u_n\|_{\mathcal{W}(0,T)} \leq C_{\text{par}} \left(\|w_n + r_n\|_{L^2(Q)} + \|u_0\|_{H_0^1(\Omega)} \right).$$

By (5.10),

$$\sup_{n \in \mathbb{N}} \|u_n\|_{\mathcal{W}(0,T)} < \infty.$$

Since $\mathcal{W}(0, T)$ is reflexive, there exist a subsequence, not relabeled, and $u^* \in \mathcal{W}(0, T)$ such that

$$u_n \rightharpoonup u^* \quad \text{weakly in } \mathcal{W}(0, T). \quad (5.11)$$

By the compact parabolic embedding $\mathcal{W}(0, T) \hookrightarrow C([0, T]; L^2(\Omega))$,

$$u_n \longrightarrow u^* \quad \text{strongly in } C([0, T]; L^2(\Omega)). \quad (5.12)$$

In particular,

$$u_n \longrightarrow u^* \quad \text{strongly in } L^2(Q). \quad (5.13)$$

The convergence in (5.12) and the identity $u_n(0) = u_0$ imply

$$u^*(0) = u_0.$$

Moreover, since

$$u_n \in L^2(0, T; H^2(\Omega) \cap H_0^1(\Omega))$$

and the latter is a closed linear subspace, the weak limit satisfies the homogeneous Dirichlet boundary condition. Hence $u^* \in \mathcal{A}$.

Define

$$g : \mathbb{R} \times \mathbb{R} \longrightarrow [0, \infty), \quad g(s, z) := \text{dist}^2(z, \Phi(s)).$$

By Lemma 5.2, the function $g(s, z) := \text{dist}^2(z, \Phi(s))$ is a normal convex integrand. Furthermore, for every fixed $s \in \mathbb{R}$, the map

$$z \longmapsto g(s, z)$$

is convex, since $\Phi(s)$ is closed and convex.

The linear operator

$$\mathcal{D} : \mathcal{W}(0, T) \longrightarrow L^2(Q), \quad \mathcal{D}u := \partial_t u - \Delta u,$$

is continuous. Therefore, (5.11) implies

$$\partial_t u_n - \Delta u_n \rightharpoonup \partial_t u^* - \Delta u^* \quad \text{weakly in } L^2(Q). \quad (5.14)$$

Using (5.13), (5.14), and the lower semicontinuity theorem for integral functionals associated with normal integrands that are convex in the second variable, we obtain

$$\begin{aligned} \mathcal{J}(u^*) &= \int_Q g(u^*, \partial_t u^* - \Delta u^*) \, dx \, dt \\ &\leq \liminf_{n \rightarrow \infty} \int_Q g(u_n, \partial_t u_n - \Delta u_n) \, dx \, dt \\ &= \liminf_{n \rightarrow \infty} \mathcal{J}(u_n) = 0. \end{aligned}$$

Since $\mathcal{J}$ is nonnegative, $\mathcal{J}(u^*) = 0$.

Part (a) then gives

$$\partial_t u^* - \Delta u^* \in \Phi(u^*) \quad \text{for a.e. } (t, x) \in Q.$$

Thus u^* is a strong solution of (2.4), and consequently $u^* \in \mathcal{S}(u_0)$. $\square$

6. Numerical Simulations

To assess the numerical performance of the DR-PINNs approach proposed in the preceding sections, next we consider three numerical examples: two for the case of differential inclusions and another for a parabolic partial differential inclusion.

6.1. Linear control system with polytopic input set

Problem formulation. We consider the two-dimensional linear differential inclusion

$$\dot{x}(t) \in Ax(t) + BU, \quad x(0) = x_0, \quad t \in [0, T], \quad (6.1)$$

where the admissible-velocity set is

$$F(t, x) = \text{conv}(V(x)), \quad V(x) = \{Ax + Bu_k\}_{k=1}^K. \quad (6.2)$$

This is an instance of regime (b) in Section 4.2: the vertex set $V(x)$ depends on the current state through an affine image of the fixed polytope $U = \text{conv}\{u_1, \dots, u_K\}$.

System data. The state dimension is $d = 2$ and the control dimension is $m = 2$. The drift matrix and the control-influence matrix are

$$A = \begin{pmatrix} -0.5 & 1.0 \\ -1.0 & -0.5 \end{pmatrix}, \quad B = I_2, \quad (6.3)$$

where I_2 denotes the 2×2 identity matrix. The eigenvalues of A are $-0.5 \pm i$, so the uncontrolled system is a stable spiral. The initial condition is $x_0 = (1, 0)^\top$ and the time horizon is $T = 1.5$.

Control set and Quickhull preprocessing. The control set is the ℓ^1 diamond in $\mathbb{R}^2$,

$$U = \text{conv}\{e_1, -e_1, e_2, -e_2\}, \quad (6.4)$$

where e_1, e_2 are the standard basis vectors. To illustrate the computational benefit of the Quickhull reduction, we augment the four true extreme vertices with eight additional points drawn uniformly from $[-0.35, 0.35]^2$, which lie generically in the interior of $\text{conv}(U)$. The resulting raw vertex set has $K_{\text{raw}} = 12$ points. Applying the Quickhull algorithm (Section 4.2, regime (b)) to BU_{raw} recovers the four true extreme points, reducing the QP (4.4) from $K_{\text{raw}} = 12$ to $\tilde{K} = 4$ variables (see Figure 1).

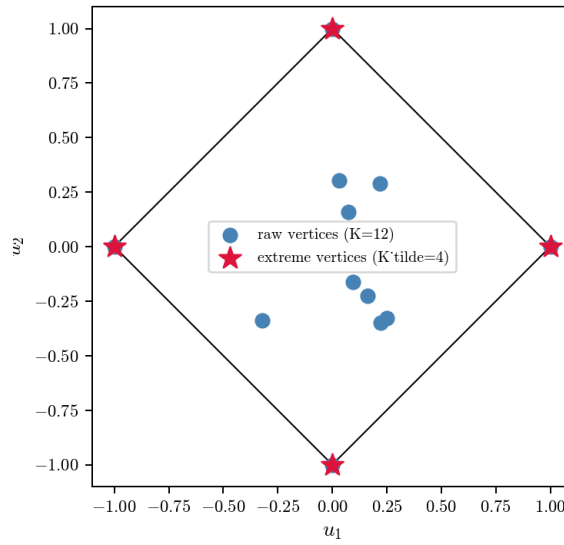


Figure 1: Raw vertex set U_{raw} ($K_{\text{raw}} = 12$, blue circles: the four true extreme vertices of the ℓ^1 diamond plus eight redundant interior points) and the four extreme vertices $\tilde{K} = 4$ recovered by Quickhull (red stars), together with the convex hull of U_{raw} .

Since $B = I_2$, the offline-reduced set is $\tilde{U} = \{e_1, -e_1, e_2, -e_2\}$, and the vertex set at each collocation point is built online as

$$\tilde{V}(x_\theta(t_i)) = \{Ax_\theta(t_i) + \tilde{u}_k\}_{k=1}^{\tilde{K}}. \quad (6.5)$$

The projection $\Pi_{F_i}(\dot{x}_\theta(t_i))$ is then computed by solving the QP (4.4) with $\tilde{K} = 4$ variables using the SLSQP method.

Neural network and training. The trial solution takes the form

$$x_\theta(t) = x_0 + t N_\theta(t), \quad (6.6)$$

where $N_\theta : \mathbb{R} \rightarrow \mathbb{R}^2$ is a fully connected neural network with tanh activation, three hidden layers of width 64, and Glorot-normal initialization. The parametrization (6.6) enforces $x_\theta(0) = x_0$ exactly for every θ , so the DR-PINN loss reduces to the pure inclusion term

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \text{dist}^2(\dot{x}_\theta(t_i), F(x_\theta(t_i))), \quad (6.7)$$

evaluated at $N = 300$ collocation points distributed uniformly on $[0, T]$. Training is performed with the Adam optimizer at learning rate 10^{-3} for 1 000 epochs. Figure 2 shows the inclusion loss $\mathcal{L}(\theta)$ of (6.7) along training. Starting from $\mathcal{L}(\theta) \approx 2.1 \times 10^{-1}$ at initialization, the loss decreases to 1.2×10^{-12} after 1 000 epochs, confirming that the learned velocity $\dot{x}_\theta(t)$ becomes admissible, i.e. $\dot{x}_\theta(t) \in F(x_\theta(t))$, at (numerically) every collocation point.

Reference reachable tube. A reference inner approximation of the reachable tube

$$\mathcal{R}(t) = \{x(t) : \dot{x}(s) = Ax(s) + Bu(s), u(s) \in U \text{ a.e.}, x(0) = x_0\} \quad (6.8)$$

is constructed by integrating the ODE $\dot{x} = Ax + Bu$ with two families of admissible controls:

- (a) *Constant bang-bang controls:* one trajectory per extreme vertex $u_k \in \tilde{U}$, totalling four reference trajectories. By Pontryagin's maximum principle, since the Hamiltonian of a linear system is affine in the control, its maximum over the polytope U is attained at an extreme point; hence the boundary of $\mathcal{R}(t)$ is traced by bang-bang controls taking values in $\tilde{U}$ (see, e.g., [20], Chapter 4, or [9], Chapter 8). This family covers the boundary of $\mathcal{R}(t)$.

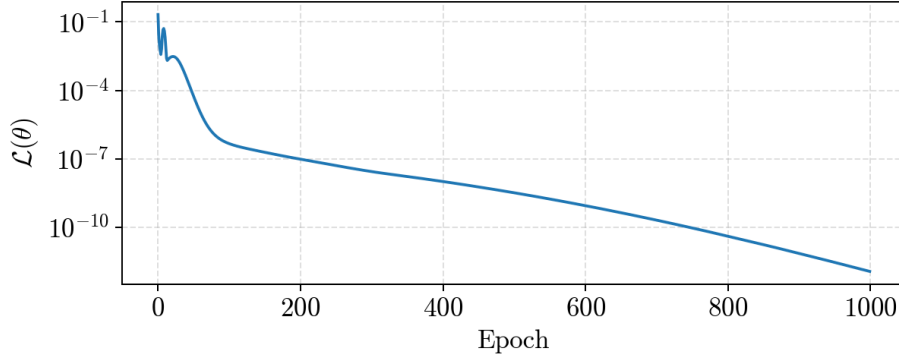


Figure 2: Training history of the inclusion loss $\mathcal{L}(\theta)$ for the single DR-PINN of this section (semi-logarithmic scale).

- (b) *Piecewise-constant random controls*: 300 trajectories obtained by randomly switching among the vertices of U at uniformly distributed times, each trajectory integrated with a fifth-order Runge–Kutta scheme (relative tolerance 10^{-9} , absolute tolerance 10^{-11}). This family densely samples the interior of $\mathcal{R}(t)$.

The union of all 304 trajectories evaluated on a uniform time grid of 200 points constitutes the reference tube used for comparison.

Figure 3 overlays the trained trajectory $x_\theta(t)$ with the reference reachable tube $\mathcal{R}(t)$ of (6.8): the left panel shows the phase portrait, the right panel the two state components against time. The learned trajectory remains inside the reference tube throughout $[0, T]$, an empirical illustration of Theorem 4.4: a vanishing inclusion loss produces an admissible trajectory.

Ensemble and Hausdorff metric. Since a single DR-PINN approximates one element of the solution set $\mathcal{S}(x_0)$ (cf. Section 4.3), its distance to the reference tube is not, by itself, a robust check: we train an ensemble of $n_{\text{ens}} = 5$ networks with different random seeds and average the resulting metric over the ensemble. The PINN reachable tube at time t is defined as

$$\mathcal{R}_\theta(t) = \{x_\theta^{(s)}(t) : s = 1, \dots, n_{\text{ens}}\}.$$

Admissibility is measured by the one-sided Hausdorff distance

$$d_H^+(\mathcal{R}_\theta(t), \mathcal{R}(t)) = \max_{a \in \mathcal{R}_\theta(t)} \min_{b \in \mathcal{R}(t)} \|a - b\|,$$

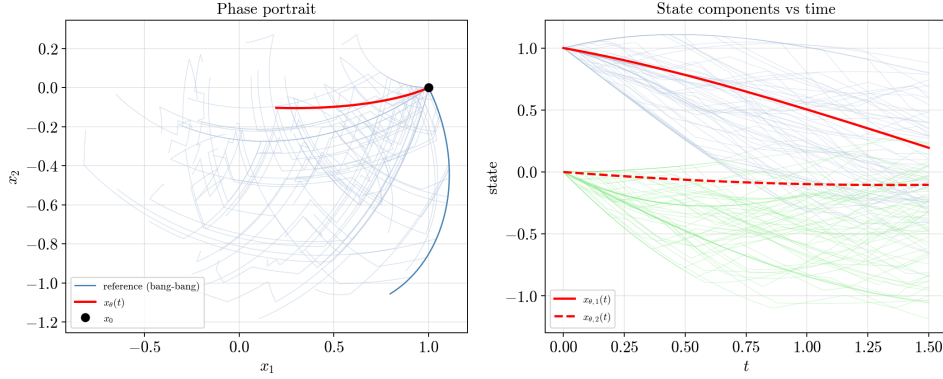


Figure 3: Phase portrait (left) and state components versus time (right) for the trained trajectory $x_{\theta}(t)$, compared with the reference reachable tube $\mathcal{R}(t)$ (304 trajectories, light blue) and one representative bang-bang extremal (dark blue).

which quantifies how far the ensemble trajectories lie from the reference tube, averaged over the ensemble and over the time grid. Averaging the one-sided Hausdorff distance $d_H^+(\mathcal{R}_{\theta}(t), \mathcal{R}(t))$ over the ensemble of $n_{\text{ens}} = 5$ networks and over the time grid gives

$$\overline{d_H^+}(\mathcal{R}_{\theta}, \mathcal{R}) = 0.0445,$$

which is small relative to the scale of the problem (the control set U and the reachable tube $\mathcal{R}(t)$ both have diameter of order 1), showing that the ensemble reliably produces admissible trajectories regardless of the random initialization.

Computational scalability. Computational efficiency is assessed by a scalability study that varies $K_{\text{raw}} \in \{8, 16, 32, 64, 128\}$ (at fixed $d = 2$) and $d \in \{2, 4, 6, 8\}$ (at fixed $K_{\text{raw}} = 32$), comparing the per-epoch training time with and without the Quickhull reduction. Table 1 lists the number of extreme vertices $\tilde{K}$ recovered by Quickhull for each raw vertex-set size $K_{\text{raw}} \in \{8, 16, 32, 64, 128\}$, at fixed $d = 2$. As in the main example, the raw point cloud consists of a fixed hexagonal boundary (six true extreme points) plus $K_{\text{raw}} - 6$ redundant interior points, so $\tilde{K} = 6$ irrespective of K_{raw} .

For the dimension sweep, a fresh random drift matrix $A_d \in \mathbb{R}^{d \times d}$ is generated for each $d \in \{2, 4, 6, 8\}$, together with a random polytope of $K_{\text{raw}} = 32$ vertices. To guarantee that the uncontrolled system $\dot{x} = A_d x$ remains asymptotically stable regardless of the random draw, A_d is constructed by first sampling a Gaussian matrix G (i.i.d. entries $\mathcal{N}(0, 0.3^2)$)

K_{raw}	8	16	32	64	128
$\tilde{K}$	6	6	6	6	6

Table 1: Extreme vertices $\tilde{K}$ recovered by Quickhull as K_{raw} grows, at fixed $d = 2$.

and then shifting its diagonal,

$$(A_d)_{ii} = G_{ii} - \left(\sum_{j=1}^d |G_{ij}| + 0.5 \right), \quad (A_d)_{ij} = G_{ij} \quad (j \neq i).$$

By the Gershgorin circle theorem, every eigenvalue λ of A_d lies in a disc centred at the real number $(A_d)_{ii}$ with radius $R_i = \sum_{j \neq i} |(A_d)_{ij}| = \sum_{j \neq i} |G_{ij}|$, so

$$\text{Re}(\lambda) \leq (A_d)_{ii} + R_i = G_{ii} - |G_{ii}| - 0.5 \leq -0.5$$

for every row i , since $G_{ii} \leq |G_{ii}|$. Hence every eigenvalue of A_d has real part at most -0.5 , for every random draw and every d : the construction guarantees a uniform stability margin without ever having to reject an unstable sample.

Table 2 shows that, unlike the K -sweep, $\tilde{K}$ now grows with d : a random point cloud in higher dimensions has a larger fraction of extreme points, and at $d = 6, 8$ all 32 raw vertices are already extreme, so no reduction is possible.

d	2	4	6	8
$\tilde{K}$	12	23	32	32

Table 2: Extreme vertices $\tilde{K}$ recovered by Quickhull as the state dimension d grows, at fixed $K_{\text{raw}} = 32$.

Figure 4 reports the resulting per-epoch training time with and without the offline Quickhull reduction. As a function of K_{raw} (left panel), the Quickhull-reduced training time stays essentially flat, between 760 and 850 ms, while the unreduced QP grows from 888 ms to 7.07 s at $K_{\text{raw}} = 128$ – an $8.3\times$ slowdown – because the QP of Section 4.2 then has 128 variables instead of 6. As a function of d (right panel), the benefit shrinks as $\tilde{K}$ approaches K_{raw} : at $d = 6$ and $d = 8$, where Table 2 shows no reduction is achieved, the Quickhull-reduced time (1.61 s, 1.48 s) is no longer lower than the unreduced time (1.56 s in both cases), since the preprocessing step then adds overhead without shrinking the downstream QP.

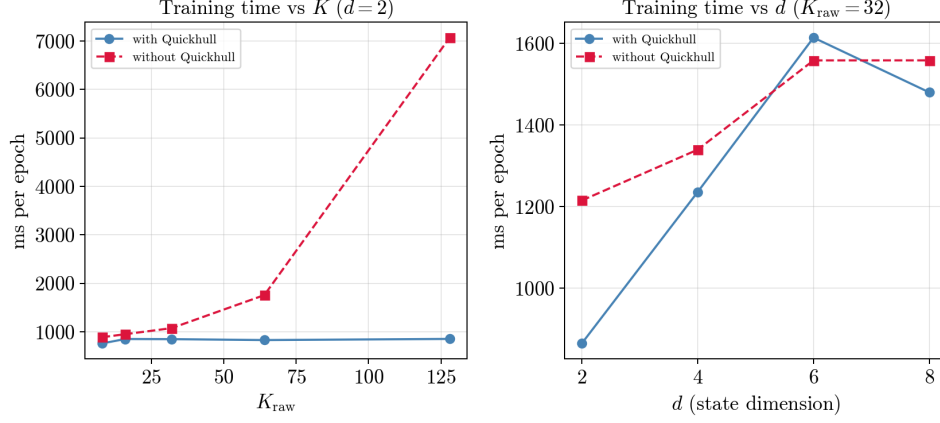


Figure 4: Per-epoch training time with (blue) and without (red) the offline Quickhull reduction. Left: as a function of K_{raw} at fixed $d = 2$ (Table 1). Right: as a function of the state dimension d at fixed $K_{\text{raw}} = 32$ (Table 2).

6.2. A Planar Differential Inclusion with a Rotating Ellipsoidal Control Constraint

Problem formulation. The two examples of Sections 6.1 and 6.3 enforce the inclusion by *minimizing* the distance residual. This second ODE example illustrates the complementary strategy anticipated in Section 4.1: when the admissible set has a tractable parametric description, admissibility can be built into the parametrization itself, so that the distance residual vanishes *identically* along every candidate trajectory and the remaining degrees of freedom can be used to select a specific element of the solution set $\mathcal{S}(x_0)$ — here, the trajectory steering the state closest to a prescribed terminal point. This makes concrete the non-uniqueness discussion of Section 4.3: the distance-residual loss guarantees convergence to *some* admissible trajectory, whereas a hard-admissible parametrization allows one to optimize *which* one.

We consider trajectories $x : [0, T] \rightarrow \mathbb{R}^2$ of the differential inclusion

$$\dot{x}(t) \in F(t, x(t)) = g(t, x(t)) + E(t), \quad x(0) = x_0, \quad (6.9)$$

whose admissible-velocity set is of Minkowski type: a nonlinear, nonautonomous drift

$$g(t, x) = \begin{pmatrix} \sin(x_1) + 0.15 \cos\left(\frac{2\pi t}{T}\right) \\ 0.5 \cos(x_2) - 0.20 \sin\left(\frac{2\pi t}{T}\right) \end{pmatrix}, \quad x = (x_1, x_2)^\top, \quad (6.10)$$

translated by a compact, convex, *time-dependent* control set $E(t)$, a rotating ellipse defined below. Equivalently, admissible trajectories are exactly the

solutions of the controlled system

$$\dot{x}(t) = g(t, x(t)) + \xi(t), \quad \xi(t) \in E(t) \text{ a.e. on } [0, T], \quad (6.11)$$

where ξ is a measurable selector of $t \mapsto E(t)$. Since $E(t)$ does not depend on the state, this example falls into regime (a) of Section 4.2; however, unlike the polytopic set of Section 6.1, the projection onto an ellipse admits no closed form (it requires the solution of a scalar secular equation), which makes the hard-admissible route particularly attractive here. We take $T = 3$ and $x_0 = (0, 0)^\top$.

The rotating ellipsoidal control set. For $t \in [0, T]$, let

$$a(t) = 0.3 + 0.4 \left| \sin\left(\frac{\pi t}{T}\right) \right|, \quad b(t) = 0.15 + 0.25 \left| \cos\left(\frac{\pi t}{T}\right) \right|, \quad \varphi(t) = \frac{\pi t}{T}, \quad (6.12)$$

and let $R_{\varphi(t)}$ denote the rotation by the angle $\varphi(t)$. The control set is the (filled) rotating ellipse

$$E(t) = \left\{ R_{\varphi(t)} \begin{pmatrix} u \\ w \end{pmatrix} \in \mathbb{R}^2 : \frac{u^2}{a(t)^2} + \frac{w^2}{b(t)^2} \leq 1 \right\}, \quad (6.13)$$

which is nonempty, compact, and convex for every t , with semi-axes and orientation varying continuously in time; since $\varphi(0) = 0$ and $\varphi(T) = \pi$, the ellipse performs a half-turn over the horizon, and the translated family $g(t, x(t)) + E(t)$ traces a twisted tube of admissible velocities along the drift curve (Figure 5). Writing $(u, w)^\top = R_{-\varphi(t)} \xi$ for the coordinates of $\xi \in \mathbb{R}^2$ in the local frame of the ellipse, the *level function*

$$\ell(t, \xi) = \left(\frac{u}{a(t)} \right)^2 + \left(\frac{w}{b(t)} \right)^2 \quad (6.14)$$

characterizes admissibility: $\xi \in E(t)$ if and only if $\ell(t, \xi) \leq 1$, with $\ell = 1$ on the boundary $\partial E(t)$.

Hard-admissible selector parametrization. The selector is generated from $n_c = 24$ control nodes $(p^j, \rho^j) \in \mathbb{R}^2 \times \mathbb{R}$, $j = 0, \dots, n_c - 1$, linearly interpolated to the uniform time grid $t_i = i \Delta t$, $i = 0, \dots, n_t - 1$, $\Delta t = T/(n_t - 1)$, with $n_t = 120$; the low-dimensional nodal representation suppresses artificial oscillations of the selector. From the interpolated values $p(t_i) \in \mathbb{R}^2$, $\rho(t_i) \in \mathbb{R}$, define the unit-ball direction and the sigmoidal radius

$$d(t_i) = \frac{p(t_i)}{\sqrt{|p(t_i)|^2 + \varepsilon}}, \quad s(t_i) = \frac{1}{1 + e^{-\rho(t_i)}} \in (0, 1), \quad (6.15)$$

with $\varepsilon = 10^{-10}$ a normalization safeguard, and set

$$\xi(t_i) = R_{\varphi(t_i)} \begin{pmatrix} s(t_i) a(t_i) d_1(t_i) \\ s(t_i) b(t_i) d_2(t_i) \end{pmatrix}. \quad (6.16)$$

Substituting (6.16) into (6.14) gives, for every grid point,

$$\ell(t_i, \xi(t_i)) = s(t_i)^2 |d(t_i)|^2 \leq s(t_i)^2 < 1, \quad (6.17)$$

so the selector lies in the interior of $E(t_i)$ *by construction*, for every value of the nodal variables: no penalty for constraint violation is required, and the inclusion residual $\text{dist}(\dot{x}(t_i) - g(t_i, x(t_i)), E(t_i))$ of Section 4.1 vanishes identically. This is the analogue, for the inclusion constraint itself, of the hard-constraint ansatz used for the initial condition in Section 6.1 and for the initial/boundary conditions in Section 6.3.

Steering problem and discretization. For a fixed selector, the state is propagated through (6.11) by the classical fourth-order Runge–Kutta scheme on the grid $\{t_i\}$, with ξ evaluated between grid points by linear interpolation. The nodal variables $z = (p^0, \dots, p^{n_c-1}, \rho^0, \dots, \rho^{n_c-1}) \in \mathbb{R}^{3n_c} = \mathbb{R}^{72}$ are chosen by minimizing

$$J(z) = \lambda_f |x_z(T) - x_f|^2 + \lambda_\xi J_\xi(z) + \lambda_s J_s(z) + \lambda_c J_c(z) + \lambda_b J_b(z) + \lambda_u J_u(z), \quad (6.18)$$

where x_z denotes the discrete trajectory induced by z , the terminal term steers the state to a prescribed target x_f , and the regularizers are the mean level $J_\xi = \frac{1}{n_t} \sum_i \ell(t_i, \xi_z(t_i))$ (discouraging boundary-hugging selectors), the mean squared first and second discrete differences of ξ_z (J_s, J_c : smoothness and curvature), the cubic soft barrier $J_b = \frac{1}{n_t} \sum_i \ell(t_i, \xi_z(t_i))^3$ (penalizing boundary saturation), and the mean squared differences of the nodal variables (J_u). The weights are $\lambda_f = 2500$, $\lambda_s = 1$, $\lambda_c = 0.2$, $\lambda_b = 10^{-2}$, $\lambda_u = 5 \cdot 10^{-2}$, and the selector-regularization weight is driven along the continuation sequence $\lambda_\xi \in \{10^{-3}, 10^{-2}, 10^{-1}\}$, each stage warm-started from the previous one and solved by L-BFGS-B.

The target must of course be chosen inside the reachable set of (6.9): the drift component $g_2 = 0.5 \cos(x_2) - 0.2 \sin(2\pi t/T)$ pushes the state upward with magnitude ≈ 0.5 near $x_2 = 0$, while the vertical reach of $E(t)$ oscillates between 0.15 and 0.7, so targets with markedly negative x_2 are unreachable over $[0, T]$ (an extremal selector taking, at every instant, the support point of $E(t)$ in the direction $-e_2$ only attains $x_2(T) \approx -0.05$). We take $x_f = (1.8, 0.1)^\top$, which lies inside but close to the boundary of the reachable set: steering to it forces the selector to operate near $\partial E(t)$ over most of the horizon, so the built-in admissibility guarantee (6.17) is genuinely stressed.

Numerical results. Table 3 reports, for each continuation stage, the terminal mismatch and the mean and maximum of the level function along the optimized selector. The terminal state matches the target to $\sim 10^{-5}$ at every stage, while increasing λ_ξ pulls the selector measurably towards the interior of the ellipse (mean level decreasing from 0.940 to 0.914) at essentially no cost in terminal accuracy — the expected trade-off for a target close to the reachable-set boundary, where near-extremal velocities are necessary for most of the horizon.

λ_ξ	$ x_z(T) - x_f $	mean ℓ	max ℓ
10^{-3}	9.0×10^{-6}	0.940	0.948
10^{-2}	6.9×10^{-6}	0.919	0.923
10^{-1}	3.2×10^{-5}	0.914	0.921

Table 3: Continuation over the selector-regularization weight λ_ξ (warm-started L-BFGS-B): terminal mismatch and level statistics of the optimized selector. Admissibility $\ell(t_i, \xi_z(t_i)) < 1$ holds at every grid point and every stage, by construction.

Figure 5 displays the resulting geometry in the space (v_1, v_2, t) of velocities and time: the translated ellipses $g(t_i, x_z(t_i)) + E(t_i)$ form a twisted tube whose centerline is the drift curve, and the selected velocity $\dot{x}_z(t_i) = g(t_i, x_z(t_i)) + \xi_z(t_i)$ runs inside the tube, close to its wall, as quantified by the level statistics of Table 3. Figure 6 shows the same information in selector coordinates: the trajectory $t \mapsto \xi_z(t)$ threads the rotating ellipse $E(t)$ centered at the origin, with the color scale reporting the level $\ell(t_i, \xi_z(t_i))$. Finally, Figure 7 shows the optimized state trajectory from x_0 to x_f (left) together with the admissibility level along the horizon (right), which stays strictly below the boundary value $\ell = 1$ at all times.

DR-PINN companion run. The hard-admissible selector optimization above is a direct constrained optimal-control discretization, not a DR-PINN: it propagates the state by RK4 and makes the inclusion hold by construction. To test the distance-residual method itself on the same benchmark, we therefore performed a separate DR-PINN run. Because the experiment is a steering problem with prescribed initial and terminal states, we used the endpoint-hard neural ansatz

$$x_\theta(t) = x_0 + \frac{t}{T}(x_f - x_0) + t(T - t)N_\theta(t), \quad (6.19)$$

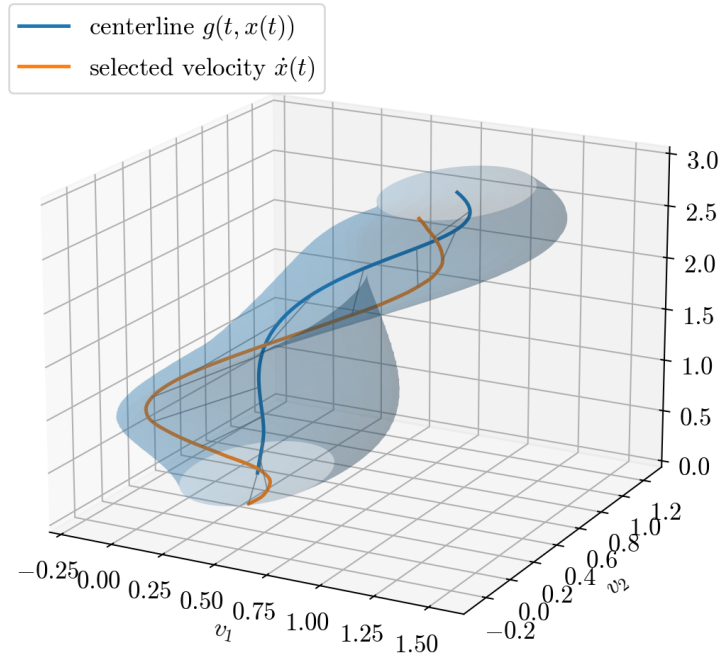


Figure 5: Admissible velocity tube of the inclusion (6.9) in (v_1, v_2, t) space. The surface is the boundary of the translated rotating ellipses $g(t, x_z(t)) + E(t)$; the blue curve is the centerline $g(t, x_z(t))$, the orange curve the selected velocity $\dot{x}_z(t)$, and the grey segments the selector displacements $\xi_z(t_i)$.

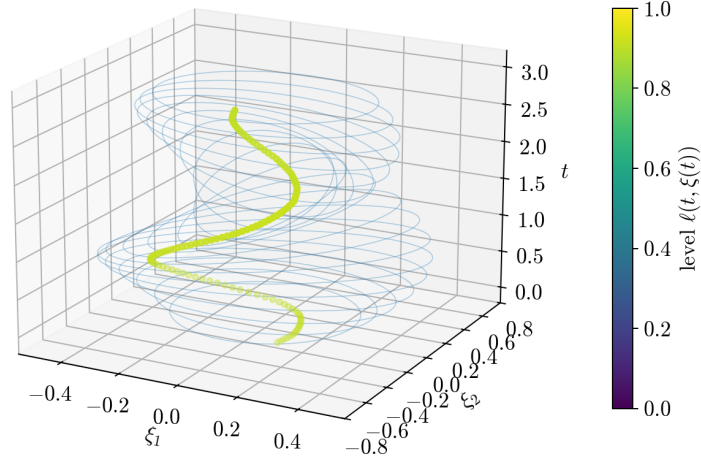


Figure 6: Optimized selector $\xi_z(t)$ inside the rotating control ellipse $E(t)$ (thin blue rings, drawn every sixth grid point). The color of the points encodes the level $\ell(t_i, \xi_z(t_i))$, which remains strictly below 1 by construction.

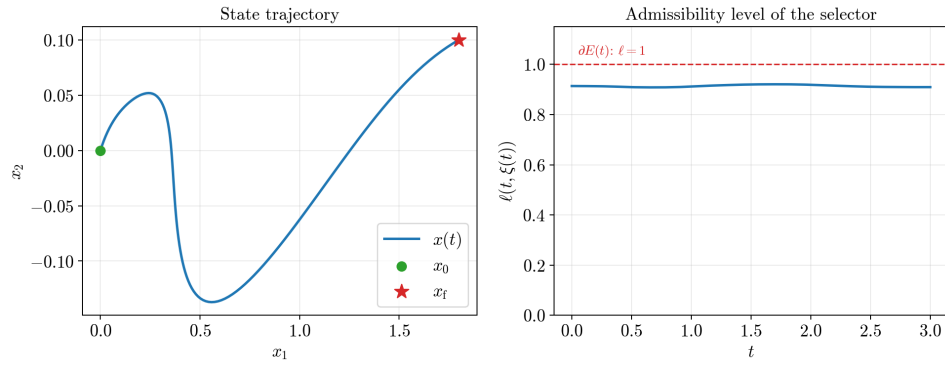


Figure 7: Left: optimized state trajectory of (6.11) from $x_0 = (0, 0)^\top$ to the target $x_f = (1.8, 0.1)^\top$ (final continuation stage $\lambda_\xi = 10^{-1}$, terminal mismatch 3.2×10^{-5}). Right: admissibility level $\ell(t, \xi_z(t))$ along the horizon; the dashed line marks the boundary value $\ell = 1$ of $\partial E(t)$.

where N_θ is a 3×64 tanh network. Hence $x_\theta(0) = x_0$ and $x_\theta(T) = x_f$ exactly, and the training objective is the pure distance residual

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \text{dist}^2(\dot{x}_\theta(t_i) - g(t_i, x_\theta(t_i)), E(t_i)), \quad (6.20)$$

with $N = 512$ uniform collocation points.

The projection onto an ellipse is evaluated pointwise by Newton iteration on the secular equation $\sum_k (\sigma_k v_k^{\text{loc}} / (\sigma_k^2 + \mu))^2 = 1$ in the local frame of $E(t)$, with $\sigma_1 = a(t)$ and $\sigma_2 = b(t)$. In the backward pass the *entire projected point* $\Pi_{E(t)}(v)$ is held constant, so automatic differentiation returns the exact envelope gradient $2(v - \Pi_{E(t)}(v))$. Holding only the multiplier μ constant would not give this gradient; this distinction is important in an implementation with a numerically computed projection. A finite-difference gradient check in the replication code gives a relative error of 7.5×10^{-11} .

After 7000 Adam epochs (hold-then-decay learning rate, double precision), the collocation loss is 1.9×10^{-14} . On an independent dense grid of 2000 time points, the mean and maximum inclusion distances are respectively 7.1×10^{-9} and 2.2×10^{-6} , while both endpoint errors are zero up to machine precision. The corresponding selector has mean level 0.931 and maximum level 1.000011; the latter tiny overshoot is consistent with the reported 2.2×10^{-6} distance and numerical projection tolerance. Thus the corrected distance-residual implementation does solve this benchmark as a genuine DR-PINN. The hard-admissible selector experiment remains useful as a complementary construction that guarantees feasibility independently of training, but it should not itself be labelled a DR-PINN experiment.

6.3. A Reaction-Diffusion Inclusion with Relay (Sliding-Mode) Feedback

As a benchmark for the parabolic setting of Section 5, we consider the reaction-diffusion inclusion

$$\partial_t u(t, x) - \Delta u(t, x) \in \Phi(u(t, x)), \quad (t, x) \in Q = (0, T) \times \Omega, \quad (6.21)$$

with homogeneous Dirichlet boundary condition, where the set-valued reaction term is the relay (signum) multifunction

$$\Phi(s) := -\lambda \text{Sign}(s) = \begin{cases} \{-\lambda\}, & s > 0, \\ [-\lambda, \lambda], & s = 0, \\ \{\lambda\}, & s < 0, \end{cases} \quad \lambda > 0. \quad (6.22)$$

Physically, (6.21)–(6.22) is the parabolic analogue of the bang-bang differential inclusion of Section 6.1. Indeed, at every point (t, x) , instead of a smooth reaction term, the state is driven towards the equilibrium $u \equiv 0$ by a switching (relay) feedback of constant magnitude λ , exactly as a sliding-mode controller drives a state to a manifold at constant rate regardless of its distance to that manifold. Equivalently, $\Phi = -\lambda \partial|\cdot|$, so that (6.21) may be read as the abstract evolution inclusion

$$\partial_t u \in -\partial E(u), \quad E(u) = \frac{1}{2} \|\nabla u\|_{L^2(\Omega)}^2 + \lambda \|u\|_{L^1(\Omega)}, \quad (6.23)$$

associated with the Dirichlet energy regularized by an L^1 (total-mass) penalty, which connects this example to sparsity-promoting flows in variational image processing and signal denoising.

We take $\Omega = (0, 1)^2$ and

$$u_0(x, y) = 16x(1-x)y(1-y), \quad (6.24)$$

which is smooth, nonnegative, and satisfies $u_0 \in H_0^1(\Omega)$, and take $\lambda = 0.6$ as a baseline value, together with a primary comparison sweep $\lambda \in \{0.4, 0.6, 0.8\}$ to illustrate how the extinction time (see below) decreases as λ increases. The time horizon is fixed at $T = 0.3$, which is large enough that all three relay cases have visibly reached extinction while the $\lambda = 0$ (classical heat equation) case has not, making the qualitative contrast visible in a single plot (see Figure 8 below).

We check that Φ in (6.22) satisfies (P1)–(P3) of Section 5.2, so that Theorem 5.3 applies to this example.

- (P1) For every $s \in \mathbb{R}$, $\Phi(s)$ is a singleton or the closed interval $[-\lambda, \lambda]$, hence nonempty, closed, and convex.
- (P2) Φ is upper semicontinuous: it is locally constant on $\mathbb{R} \setminus \{0\}$, and at $s = 0$, $\Phi(s_n) \subset [-\lambda, \lambda] = \Phi(0)$ for every sequence $s_n \rightarrow 0$. This is the standard Filippov regularization of the signum function [21].
- (P3) $\sup_{r \in \Phi(s)} |r| = \lambda \leq \lambda(1 + |s|)$ for every $s \in \mathbb{R}$, so (P3) holds with $m_0 = \lambda$.

Remark 6.1. This example lies outside the scope of the existence theory of Theorem 2.11, which requires a *continuous* selection $f(s) \in \Phi(s)$ (hypothesis (H2')). Indeed, any selection of Φ in (6.22) is forced to equal $-\lambda$ for $s > 0$ and λ for $s < 0$, hence is necessarily discontinuous at $s = 0$. Existence of a strong solution is instead obtained from the classical theory of evolution

equations governed by maximal monotone operators [22]: rewriting (6.21) as $\partial_t u + (-\Delta + \lambda \partial|\cdot|)(u) \ni 0$, the operator $-\Delta + \lambda \partial|\cdot|$ is a sum of two maximal monotone (subdifferential) operators on $L^2(\Omega)$.

Next, we address the finite-time extinction issue. Multiplying (6.21) by $u(t, \cdot)$, integrating over Ω , and using that $s \cdot v = -\lambda|s|$ for every $v \in \Phi(s)$ and every $s \in \mathbb{R}$, a formal computation (rigorous for strong solutions) gives the energy identity

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_{L^2(\Omega)}^2 + \|\nabla u(t)\|_{L^2(\Omega)}^2 = -\lambda \|u(t)\|_{L^1(\Omega)}. \quad (6.25)$$

The extra dissipation term $-\lambda \|u(t)\|_{L^1(\Omega)}$, absent in the classical heat equation ($\lambda = 0$), is responsible for a qualitatively different phenomenon: solutions of gradient flows of linear-growth functionals such as (6.23) are known to reach their equilibrium in *finite time* rather than merely decaying exponentially, see [23]. Once $u(t, \cdot)$ vanishes on part of Ω , the relay may “slide” there ($\Delta u(t, x) \in [-\lambda, \lambda]$ pointwise, consistently with $u \equiv 0$), anchoring the solution at equilibrium. This behavior is invisible to a classical pointwise-residual PINN, which would require committing in advance to one branch of Φ , and is precisely the phenomenon this benchmark is designed to showcase. Identity (6.25) also provides a solver-independent numerical check: the decay of $\|u_\theta(t)\|_{L^2(\Omega)}^2$ along the trained trajectory should be strictly faster than that of the corresponding heat equation ($\lambda = 0$) with the same initial datum, and the extinction time should decrease as λ increases.

Adaptive collocation sampling. Because Φ has a genuine jump discontinuity at $s = 0$, the distance residual is hardest to drive to zero in a neighborhood of the (a priori unknown, time-dependent) extinction front

$$\{(t, x) : u_\theta(t, x) = 0\}.$$

This is precisely where a smooth network output must approximate a discontinuous right-hand side, and our experiments confirm that the residual after training is concentrated there rather than spread uniformly over Q . Purely uniform collocation under-resolves this thin, moving region relative to its share of the loss. We therefore complement uniform collocation with a residual/state-based adaptive refinement scheme in the spirit of the RAR method of Lu, Meng, Mao, and Karniadakis [24]. Precisely, at fixed intervals during training, a large pool of candidate points is drawn uniformly over Q , the points with the smallest $|u_\theta|$ are retained as an auxiliary “hard pool” tracking the current estimate of the extinction front, and a fraction of every

subsequent training batch is drawn from this pool — refreshed periodically as the front moves — while the remaining points continue to be sampled uniformly over Q to preserve global coverage of the domain. In our experiments the batch size is $N_{\text{col}} = 2000$ collocation points per epoch; the adaptive fraction is 30% (600 points), drawn from a hard pool of $n_{\text{pool}} = 4000$ candidate points selected from a uniform draw of 20 000 points over Q by retaining those with smallest $|u_\theta|$. The hard pool is refreshed every 500 training epochs as the extinction front advances; the remaining 70% of each batch is drawn uniformly over Q to preserve global domain coverage.

Reference solution. Unlike the bang-bang example of Section 6.1, where F is genuinely multivalued and the ensemble/Hausdorff comparison of Section 6.1 is required, uniqueness of strong solutions holds here: since $-\Delta + \lambda \partial|\cdot|$ is a sum of maximal monotone operators, standard monotonicity arguments show that $\frac{d}{dt}\|u_1 - u_2\|_{L^2(\Omega)}^2 \leq 0$ for any two strong solutions with the same initial datum, so $\mathcal{S}(u_0)$ is a singleton. A single trained DR-PINN can therefore be validated directly against one reference solution, obtained by a first-order operator-splitting scheme on a uniform 49×49 interior grid with time step $\Delta t = 5 \times 10^{-4}$. Each time step alternates an implicit diffusion solve, $v = (I - \Delta t \Delta_h)^{-1} u^n$ (with the 5-point finite-difference Laplacian Δ_h and homogeneous Dirichlet boundary conditions), and an exact proximal step,

$$u^{n+1} = \text{sign}(v) \max(|v| - \lambda \Delta t, 0),$$

the pointwise proximal operator of $s \mapsto \lambda \Delta t |s|$. This scheme captures finite-time extinction exactly: any grid point for which $|v_i| \leq \lambda \Delta t$ is sent to identically zero by the soft-thresholding step.

Network architecture and loss function. The trial solution takes the hard-constrained form

$$u_\theta(t, x, y) = u_0(x, y) + t x(1 - x) y(1 - y) N_\theta(t, x, y), \quad (6.26)$$

where $N_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a fully connected neural network. The parametrization (6.26) enforces the initial condition $u_\theta(0, \cdot) = u_0$ and the homogeneous Dirichlet boundary condition exactly for every θ (the polynomial factor vanishes on $\partial\Omega$), so, as in Section 6.1, the DR-PINN loss reduces to the pure inclusion term: it is (5.1) with Φ as in (6.22) and $\lambda_0 = \lambda_b = 0$. Since $\Phi(s)$ is an interval of $\mathbb{R}$, the projection $\Pi_{\Phi(u_\theta(t_i, x_i))}$ needed to evaluate the distance residual is explicit, requiring no quadratic program. Beyond the usual comparison against the reference solver, we recommend a diagnostic figure specific to this set-valued setting: superimposing, on a plot of

the graph of Φ (a three-branch “staircase”), the empirical cloud of points $(u_\theta(t_i, x_i), \partial_t u_\theta(t_i, x_i) - \Delta u_\theta(t_i, x_i))$ at the collocation points after training. A well-trained network should concentrate this cloud on the graph of Φ , visually certifying that the network has learned to select the correct branch of the relay at each collocation point — a diagnostic with no counterpart in classical single-valued PINNs.

Numerical results. Figure 8 displays the $L^2(\Omega)$ -norm decay curves for the reference solver with $\lambda \in \{0, 0.4, 0.6, 0.8\}$ over $[0, T]$. The contrast is sharp: the classical heat equation ($\lambda = 0$) decays exponentially and remains well above zero at $t = T = 0.3$, while all three relay cases reach exact numerical extinction within $[0, 0.19]$, with the extinction time decreasing monotonically as λ increases. Table 4 records the reference extinction times and those detected by the trained DR-PINNs.

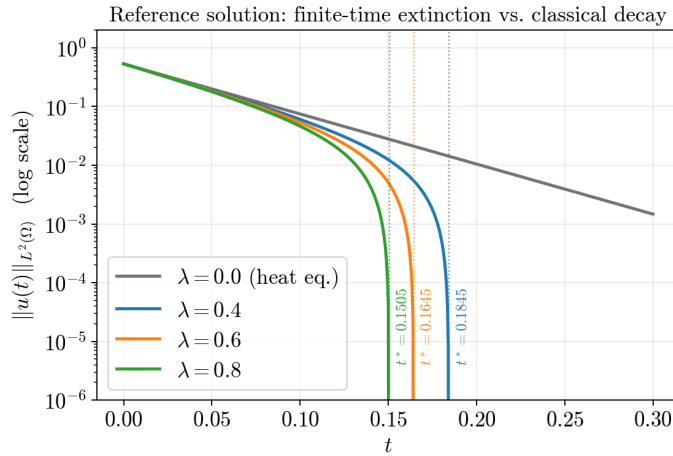


Figure 8: Reference solution: $\|u(t)\|_{L^2(\Omega)}$ on a log scale for $\lambda \in \{0, 0.4, 0.6, 0.8\}$. The vertical dotted lines mark the finite-time extinction instants t^* of the three relay cases; the $\lambda = 0$ heat-equation curve does not extinguish within $[0, T]$. Curves are clipped below 10^{-6} for readability; after extinction the reference norm is exactly zero (drop below the axis).

The network for the baseline case $\lambda = 0.6$ is a fully connected tanh network with five hidden layers of width 128 and Glorot-normal initialization, trained for 40 000 epochs with the Adam optimizer under a hold-then-decay learning-rate schedule: the rate is held at 2×10^{-3} for the first 5 000 epochs and then decayed exponentially with rate 0.85 per 3 000 steps. The same architecture and schedule are used for each λ in the primary sweep.

λ	t_{ref}^*	$t_{\text{DR-PINN}}^*$	relative error
0.4	0.1845	0.1900	3.0%
0.6	0.1645	0.1725	4.9%
0.8	0.1505	0.1625	8.0%

Table 4: Reference and DR-PINN extinction times t^* for the primary sweep $\lambda \in \{0.4, 0.6, 0.8\}$. The DR-PINN extinction time is defined as the first evaluation instant at which $\|u_\theta(t)\|_{L^2(\Omega)} < 10^{-3}$.

Figure 9 shows the training and validation loss curves for the baseline run ($\lambda = 0.6$). After a brief initial plateau at $\mathcal{L} \approx 20$ (epochs 0–4 500), the loss drops abruptly by roughly two orders of magnitude and then decreases steadily for the remainder of training, reaching a final validation loss of 3.9×10^{-1} . Training and validation curves track each other closely throughout, indicating no overfitting.

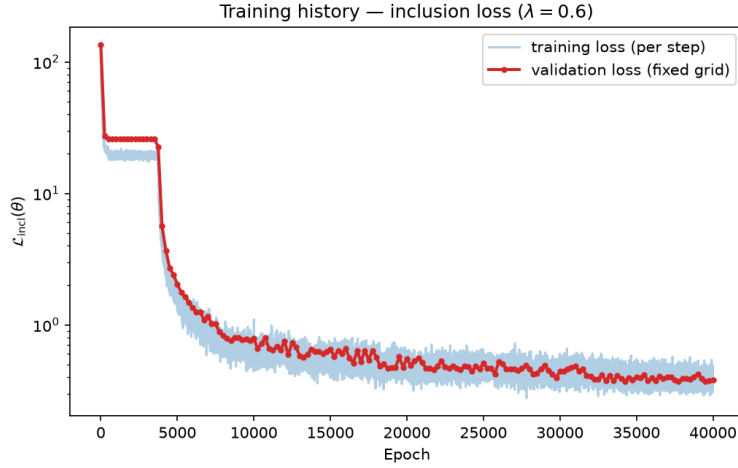


Figure 9: Training history of the inclusion loss $\mathcal{L}_{\text{incl}}(\theta)$ for the baseline DR-PINN ($\lambda = 0.6$, 40 000 epochs). The training loss per step (light blue) and the validation loss evaluated on a fixed batch of 4 000 points (red markers, every 250 epochs) track each other closely; the abrupt drop near epoch 4 500 marks the escape from the initial plateau.

Figure 10 compares the DR-PINN against the reference solution for $\lambda = 0.6$. The left panel overlays the $L^2(\Omega)$ -norm decay curves: the two coincide to within a line width for $t \lesssim 0.16$, with a maximum relative discrepancy of 0.76% over $[0, T]$. After the reference solution extinguishes at $t_{\text{ref}}^* = 0.1645$ (dropping to machine zero), the DR-PINN norm stabilizes at $\approx 10^{-4}$, since a

smooth network cannot represent an identically zero function after a sharp front. The right panel shows the pointwise absolute error $|u_\theta - u_{\text{ref}}|$ at $t = T = 0.3$: the error is at most 2×10^{-3} , smooth, and concentrated at the domain centre, a pattern consistent with the residual of a smooth approximation of the trivial solution $u \equiv 0$ after extinction.

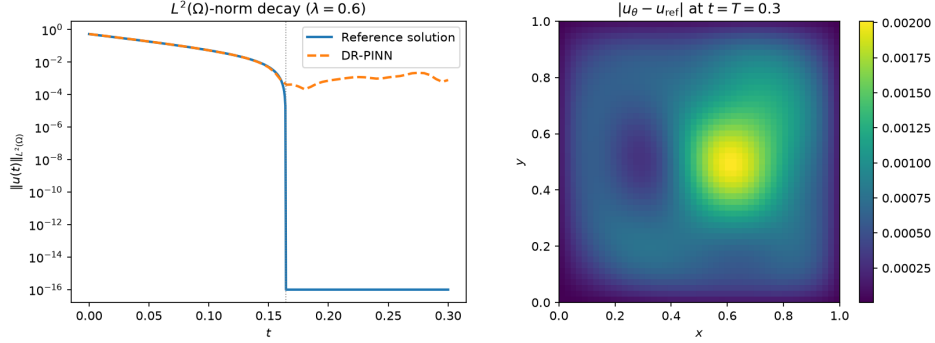


Figure 10: Validation of the DR-PINN against the reference solution ($\lambda = 0.6$). Left: $L^2(\Omega)$ -norm decay on a log scale (clipped below 10^{-6} for readability); the dotted vertical line marks the reference extinction time $t_{\text{ref}}^* = 0.1645$. Right: pointwise error $|u_\theta - u_{\text{ref}}|$ at $t = T = 0.3$; the maximum error is $\approx 2 \times 10^{-3}$.

Figure 11 shows the branch-selection diagnostic for the baseline run. For state values $s = u_\theta(t, x, y) \gtrsim 0.05$, the cloud of operator values $z = \partial_t u_\theta - \Delta u_\theta$ concentrates tightly on the branch $z = -\lambda = -0.6$ of the graph of Φ , confirming that the network has correctly selected the positive branch over most of Q . No collocation points appear with $s < 0$: since $u_0 \geq 0$ and the relay feedback pushes u towards zero without reversing sign, the exact solution satisfies $u(t, \cdot) \geq 0$ for all t , and the trained network reproduces this property implicitly. Near $s \approx 0$, corresponding to the extinction front, the cloud disperses vertically with values of z reaching up to 7, far above the segment $[-\lambda, \lambda]$ of the graph of Φ . This scatter is a direct consequence of the jump discontinuity of Φ at $s = 0$: no smooth network output can resolve a discontinuous right-hand side pointwise, and the adaptive hard pool concentrates collocation points precisely where the residual is hardest to reduce.

Figure 12 and Table 4 summarize the extinction-time comparison across the primary sweep $\lambda \in \{0.4, 0.6, 0.8\}$. The DR-PINN norm curves reproduce the qualitative shape of the reference curves for all three values of λ , and the detected extinction times follow the correct ordering $t^*(0.4) > t^*(0.6) > t^*(0.8)$. The absolute errors range from 0.006 to 0.012 time units, growing

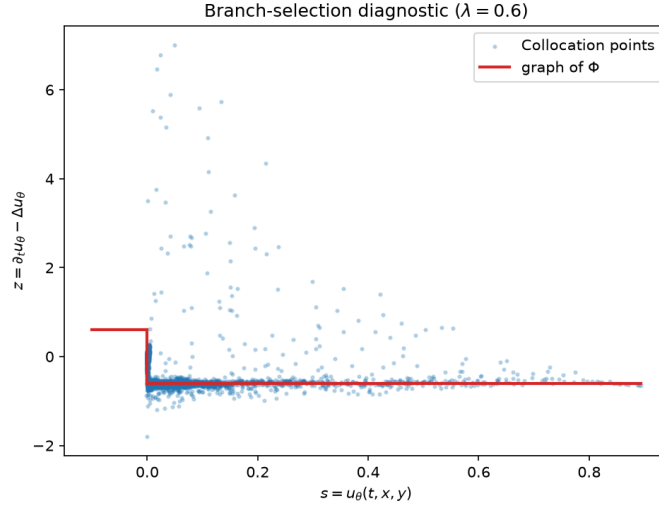


Figure 11: Branch-selection diagnostic ($\lambda = 0.6$): empirical cloud of $(u_\theta, \partial_t u_\theta - \Delta u_\theta)$ at 4000 post-training collocation points (blue dots), overlaid on the graph of Φ (red). Points with $s \gtrsim 0.05$ cluster tightly on the branch $z = -0.6$; no points appear with $s < 0$ (sign preservation by the relay flow); the vertical scatter near $s \approx 0$ traces the extinction front, where the discontinuity of Φ limits pointwise accuracy. The inset magnifies the neighborhood of the extinction front (s close to 0).

modestly with λ ; this is consistent with the observation that the effective loss scale grows as λ^2 (the distance-residual target $z \rightarrow \pm\lambda$ grows with λ), so that a fixed training budget resolves the extinction front with slightly lower absolute accuracy at larger λ .

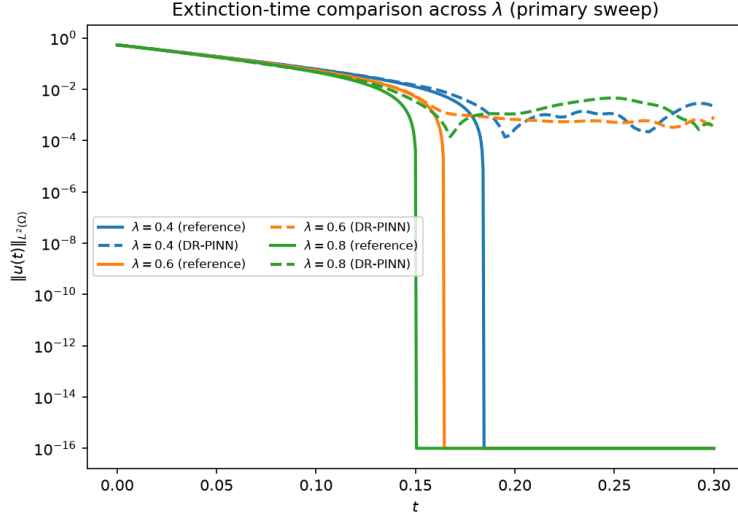


Figure 12: Extinction-time comparison across the primary sweep $\lambda \in \{0.4, 0.6, 0.8\}$. Solid lines: reference solver. Dashed lines: DR-PINN. Colors identify the three λ values (blue: 0.4, orange: 0.6, green: 0.8). Curves are clipped below 10^{-6} for readability: after extinction the reference drops to exactly zero (below the axis) while the DR-PINN norm stabilizes at $\approx 10^{-4}$; the correct monotone ordering of extinction times is reproduced by the DR-PINN for all three values.

7. Conclusions

[Conclusions to be added here.]

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